Origical Correct MTH303 Lecture Notes

Mathematical methods 2 (Obafemi Awolowo University)

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MTH 303: Advanced Calculus

December 14, 2024

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Multiple Integrals

0.1 Double Integrals

Suppose we have a region in the plane R and a function $f(x, y)$. Then the double integral

$$\iint_R f(x, y)\,\mathrm{d}A$$

is defined as follows. Divide the region R into small pieces, numbered from 1 to n. Let ΔA_i be the area of the ith piece and also pick a point (x_i, y_i) in that piece. The figure (Figure 1) shows a region R divided into small pieces and shows the ith piece with its area, and choice of a point in the little region.

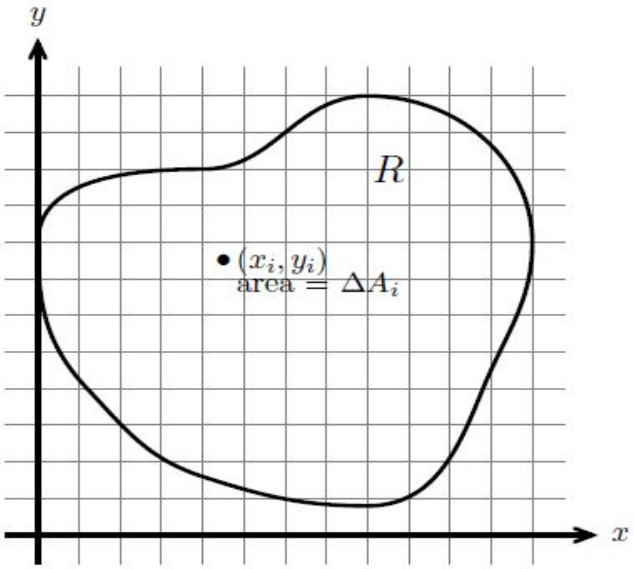


Figure 1:

Now form the sum

$$\sum_{i=1}^{n} f(x_i, y_i)\Delta A_i \tag{1}$$

and then, finally

$$\iint_R f(x, y)\,\mathrm{d}A = \lim_{\Delta A \to 0} \sum_{i=1}^{n} f(x_i, y_i)\Delta A_i.$$

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Here, the limit is taken by letting the number of pieces go to infinity and the area of each piece go to 0. There are technical requirements that the limit exist and be independent of the specific limiting process.

If $f(x,y) \geq 0$, then the volume V of the solid that lies above the rectangle R and below the surface $z = f(x,y)$ is

$$V = \iint_R f(x,y)\,\mathrm{d}A.$$

0.1.1 Properties of Double Integrals

If f and g are continuous on R and $c \in \mathbb{R}$, then

(1.) $\iint_R [f(x,y) + g(x,y)]\,\mathrm{d}A = \iint_R f(x,y)\,\mathrm{d}A + \iint_R g(x,y)\,\mathrm{d}A$

(2.) $\iint_R cf(x,y)\,\mathrm{d}A = c\iint_R f(x,y)\,\mathrm{d}A$ If $f(x,y) \geq g(x,y)$ *for all* (x,y) *in* R, then

(3.) $\iint_R f(x,y)\,\mathrm{d}A \geq \iint_R g(x,y)\,\mathrm{d}A$

(4.) If $D = D_1 U D_2$, where D_1 and D_2 don't overlap except perhaps on their boundaries, then

$$\iint_D f(x,y)\,\mathrm{d}A = \iint_{D_1} f(x,y)\,\mathrm{d}A + \iint_{D_2} f(x,y)\,\mathrm{d}A.$$

(5.) If we integrate the constant function $f(x,y) = 1$ over a region D, we get the area of D :

$$\iint_D 1\,\mathrm{d}A = A(D).$$

(6.) If $m \leq f(x,y)$ for all (x,y) in D, then

$$mA(D) \leq \iint_D 1\,\mathrm{d}A \leq MA(D).$$

EXERCISES

(a) Evaluate the following double integrals by first identifying them as the volumes of solids.
(i) $\iint_R 3\,\mathrm{d}A,\ R = \{(x,y)|\ -2 \leq x \leq 2,\ 1 \leq y \leq 6\}$, (ii) $\iint_R (5-x)\,\mathrm{d}A, \quad R = \{(x,y)| \quad 0 \leq x \leq 5, \quad 0 \leq y \leq 3\}$, (iii) $\iint_R (4-2y)\,\mathrm{d}A, \quad R = [0,1] \times [0,1]$.

(b) If f is a constant function, $f(x,y) = k$ and $R = [a,b] \times [c,d]$, show that $\iint_R k\,\mathrm{d}A = k(b-a)(d-c)$

(c) Use the result of question b to show that $0 \leq \iint_R \sin(\pi x)\cos(\pi y)\,\mathrm{d}A \leq \frac{1}{32}$, where $R = \left[0, \frac{1}{4}\right] \times \left[\frac{1}{4}, \frac{1}{2}\right]$.

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0.1.2 Iterated Integrals

Suppose that f is a function of two variables that is integrable on the rectangle $R = [a, b] \times [c, d]$. If $\int_c^d f(x, y)dy$ is a number that depends on the value of x, then a function of x is defined as

$$A(x) = \int_c^d f(x, y)dy.$$

Integrating the function A with respect to x from $x = a$ to $x = b$, we have

$$\int_a^b Adx = \int_a^b \left[\int_c^d f(x, y)dy \right] dx \tag{2}$$

The integral on the right side of Equation (2) is called an **iterated integral.** Usually the brackets are omitted and thus

$$\int_a^b \int_c^d f(x, y)dydx = \int_a^b \left[\int_c^d f(x, y)dy \right] dx \tag{3}$$

means that we first integrate with respect to y from c to d and then with respect to from a to b.
Similarly, the iterated integral

$$\int_c^d \int_a^b f(x, y)dxdy = \int_c^d \left[\int_a^b f(x, y)dx \right] dy \tag{4}$$

means that we first integrate with respect to x (holding y fixed) from $x = a$ to $x = b$ and then we integrate the resulting function of y from $y = c$ to $y = d$.
Example 1. *Evaluate the iterated integrals.*

$$(i) \quad \int_0^3 \int_1^2 x^2 y dy dx, \quad (ii) \quad \int_1^2 \int_0^3 x^2 y dx dy$$

Solution:
(i) Regarding x as a constant, we obtain

$$\int_0^3 \int_1^2 x^2 y dy dx = \int_0^3 \left[\int_1^2 x^2 y dy \right] dx$$

$$= \int_0^3 \frac{3}{2} x^2 dx = \frac{x^3}{2} \Big|_0^3 = \frac{27}{2}$$

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(ii) Here we first integrate with respect to x

$$\int_1^2 \int_0^3 x^2 y dx dy = \int_1^2 \left[\int_0^3 x^2 y dx \right] dy = \int_1^2 \left[\frac{x^3}{3} y \right] \Big|_{x=0}^{x=3} dy$$

$$= \int_1^2 9y dy = 9\frac{y^2}{2} \Big|_1^2 = \frac{27}{2}.$$

Example 2. *Find the volume of solid S that is bounded by the elliptic paraboloid $x^2 + 2y^2 + z = 16$, the plane $x = 2$, $y = 2$, and the three coordinate planes.*

Solution: *S is the solid that lies under the surface $z = 16 - x^2 - 2y^2$ and above the square $R = [0, 2] \times [0, 2]$.*

$$\iint_R \big(16 - x^2 - 2y^2\big) dA = \int_0^2 \int_0^2 \big(16 - x^2 - 2y^2\big) dx dy$$

$$= \int_0^2 \left[16x - \frac{1}{3}x^3 - 2y^2 x \right] \Big|_{x=0}^{x=2} dy$$

$$= \int_0^2 \left(\frac{88}{3} - 4y^2 \right) dy = \left[\frac{88}{3} y - \frac{4}{3} y^3 \right] \Big|_0^2 = 48$$

Suppose that $f(x, y) = g(x)h(y)$ and $R = [a, b] \times [c, d]$, then the double integral can be written as a product of two single integrals:

$$V = \iint_R g(x)h(y)\, \mathrm{d}A = \int_b^b g(x)dx \int_c^d h(y)dy,$$

where $R = [a, b] \times [c, d]$.

EXERCISES

(a) Calculate the iterated integral.
(i) $\int_{-3}^{3} \int_0^{\frac{\pi}{2}} \big(y + y^2 \cos(x)\big) dx dy$, (ii) $\int_1^3 \int_5^5 \frac{\ln y}{xy} dy dx$

(b) Calculate the double integral.
(i) $\iint_R \frac{xy^2}{x^2+1}\, \mathrm{d}A$, $R = \{(x, y)| \quad 0 \le x \le 1, -3 \le x \le 3\}$ (ii) $\iint_R x \sin(x+y)\, \mathrm{d}A$, $R = [0, \pi/6] \times [0, \pi/3]$

(c) Find the volume of the solid that lies under the plane $4x + 6y - 2z + 15 = 0$ and above the rectangle $R = \{(x, y)| -1 \le x \le 2, -1 \le x \le 1\}$

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(d) Find the volume of the solid lying under the elliptic paraboloid $x^2/4 + y^2/9 + z = 1$ and above the rectangle $R = [-1, 1] \times [-2, 2]$

Example 3. *Evaluate*

$$\iint_D (x + 2y)\,\mathrm{d}A,$$

where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Solution: The parabolas intersect when $2x^2 = 1 + x^2$, that is, $x^2 = 1$, so $x = \pm 1$

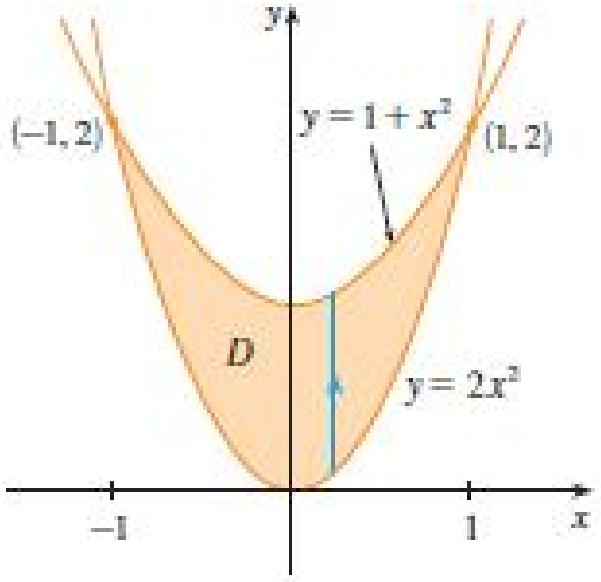


Figure 2:

This is a type I region (**Figure** 2) but not a type II region and we can write

$$D = \{(x, y)|\ -1 \le x \le 1,\ 2x^2 \le y \le 1 + x^2\}. \tag{5}$$

Since the lower boundary is $y = 2x^2$ and the upper boundary is $y = 1 + x^2$, therefore we have

$$\iint_D (x + 2y)\,\mathrm{d}A = \int_{-1}^{1} \int_{2x^2}^{1+x^2} (x + 2y)dydx$$

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$$= \int_{-1}^{1} \left[xy + y^2 \right]_{y=2x^2}^{y=1+x^2} dx$$

$$= \int_{-1}^{1} (-3x^4 - x^3 + 2x^2 + x + 1) = \frac{32}{15}.$$

Example 4. *Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region D in the xy-plane bounded by the line $y = 2x$ and the parabola $y = x^2$.*

Solution 1;

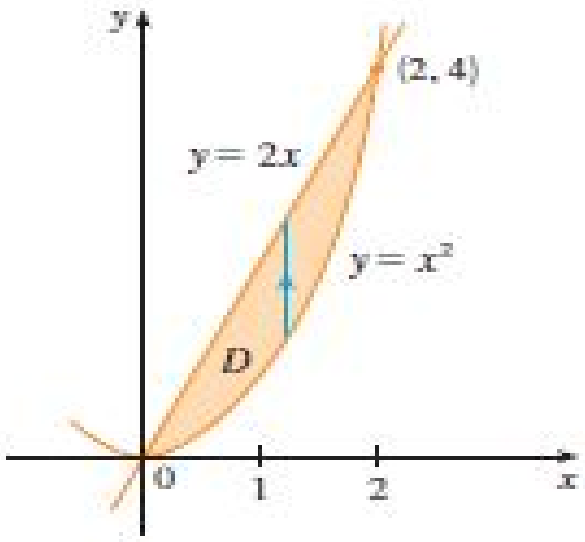


Figure 3:

From **Figure 3,** we see that D is a type I region and

$$D = \{(x, y) | \, 0 \leq x \leq 2, \; x^2 \leq y \leq 2x\}.$$

Therefore the volume under $z = x^2 + y^2$ and above D is

$$V = \iint_D (x^2 + y^2) \, \mathrm{d}A = \int_0^2 \int_{x^2}^{2x} (x^2 + y^2) dy dx$$

$$= \int_0^2 \left[x^2 y + \frac{y^3}{3} \right]_{y=x^2}^{y=2x} dx = \int_0^2 \left(-\frac{x^6}{3} - x^4 + \frac{14x^3}{3} \right) dx = \frac{216}{35}.$$

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Solution 2:

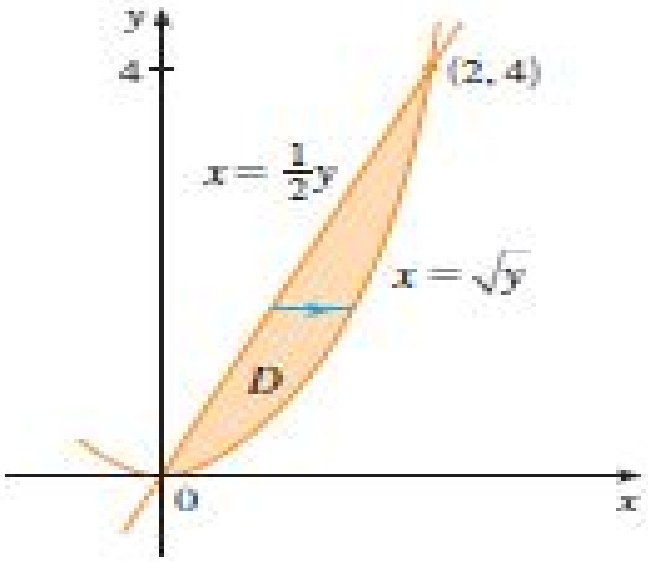


Figure 4:

Here D is a type II region and thus we have (**Figure 4**)

$$D = \Big\{(x, y)|\, 0 \le y \le 4,\ \frac{1}{2}y \le x \le \sqrt{y}\Big\}.$$

Therefore

$$V = \iint_D (x^2 + y^2)\, \mathrm{d}A = \int_0^4 \int_{\frac{1}{2}y}^{\sqrt{y}} \Big(x^2 + y^2\Big) dxdy$$

$$= \int_0^4 \left(-\frac{y^{3/2}}{3} + y^{5/2} - \frac{y^3}{24} - \frac{y^3}{2} \right) dy = \frac{216}{35}.$$

EXERCISES

(a) Find the volume of the given solid.

(i) Under the plane $x - 2y + z = 1$ and above the region bounded by $x + y = 1$ and $x^2 + y = 1$.

(ii) Under the surface $z = xy$ above the triangle with vertices $(1, 1)$, $(4, 1)$, *and* $(1, 2)$.

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(iii) Bounded by the cylinder $x^2 + y^2 = 1$ and the planes $y = z$, $x = 0$, $z = 0$ in the first octant.

(b) Find the volume of the solid by subtracting two volumes.

(i) The solid enclosed by the parabolic cylinders $y = 1 - x^2$, $y = x^2 - 1$, and the planes $x + y + z = 2$, $2x + 2y - z + 10 = 0$.

(ii) The solid enclosed by the parabolic cylinder $y = x^2$ and the planes $z = 3y$, $z = 2 + y$.

(c) Use Property 5 to estimate the value of the integral.

(i)
$$\iint_Q e^{-(x^2+y^2)^2}\,\mathrm{d}A,$$
Q is the quarter-circle with center the origin and radius $\frac{1}{2}$ in the first quadrant

(ii)
$$\iint_T \sin^4(x+y)\,\mathrm{d}A,$$
T is the triangle enclosed by the lines $y = 0$, $y = 2x$, *and* $x = 1$.

0.1.3 Double Integrals in Polar Coordinates

Double integrals in (x, y) coordinates which are taken over circular regions, or have integrands involving the combination $x^2 + y^2$, are often better done in polar coordinates:

$$\iint_R f(x,y)\,\mathrm{d}A = \iint_R g(r,\theta)\,\mathrm{d}r\mathrm{d}\theta$$

This involves introducing the new variables r and θ, together with the equations relating them to x, y in both the forward and backward directions:

$$r = \sqrt{x^2+y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right), \quad x = r\cos\theta, \quad y = r\sin\theta.$$

Thus, if f is continuous on a polar rectangle R given by $0 \le a \le r \le b$, $\alpha \le \theta \le \beta$, *where* $0 \le \beta - \alpha \le 2\pi$, then

$$\iint_R f(x,y)\,\mathrm{d}A = \int_\alpha^\beta \int_a^b f\Big(r\cos\theta, r\sin\theta\Big) r dr d\theta$$

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Example 5. *Evaluate*

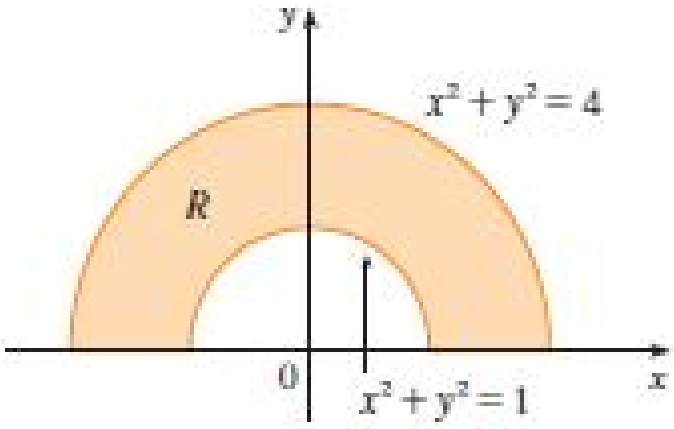
$$\iint_R (3x + 4y^2)\,\mathrm{d}A,$$

where R is the region in the upper half-plane bounded by the circle $x^2+y^2 = 1$ and $x^2+y^2 = 4$.

Solution: The region R can be described as

$$R = \{(x, y)|\ \ y \geq 0,\ 1 \leq x^2 + y^2 \leq 4\}.$$

It is the half-ring shown in **Figure 5**.



(b) $R = \{(r, \theta) \mid 1 \leq r \leq 2, 0 \leq \theta \leq \pi\}$

Figure 5:

In polar coordinates, it is given by $1 \leq r \leq 2,\ 0 \leq \theta \leq \pi$. Therefore we have

$$\iint_R \left(3x + 4y^2\right)\mathrm{d}A = \int_0^{\pi}\int_1^2 \left(3r\cos\theta + 4r^2\sin^2\theta\right) r dr d\theta$$

$$= \int_0^{\pi}\int_1^2 \left(3r^2\cos\theta + 4r^3\sin^2\theta\right) dr d\theta$$

$$= \int_0^{\pi} \left(r^3\cos\theta + r^4\sin^2\theta\right)\Big|_{r=1}^{r=2} d\theta = \int_0^{\pi} \left(7\cos\theta + 15 sin^2\theta\right) d\theta$$

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$$= \int_0^{\pi} \Big[7\cos\theta + \frac{15}{2}\Big(1 - \cos 2\theta\Big) \Big] d\theta$$

$$= 7\sin\theta + \frac{15\theta}{2} - \frac{15}{4}\sin 2\theta \Big|_0^{\pi} = \frac{15\pi}{2}.$$

NOTE: If f is continuous on a polar region of the form

$$D = \{(r,\theta) |\, \alpha \le \theta \le \beta,\ h_1(\theta) \le r \le h_2(\theta)\}$$

then

$$\iint_D f(x,y)\, \mathrm{d}A = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f\Big(r\cos\theta, r\sin\theta\Big) r\, dr\, d\theta.$$

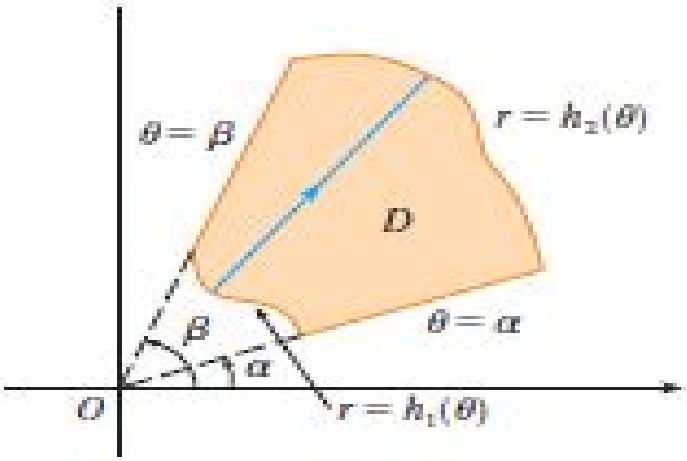


Figure 6:

Example 6. *Use a double integral to find the area enclosed by one loop of the four-leaved rose* $r = \cos 2\theta$.

Solution:

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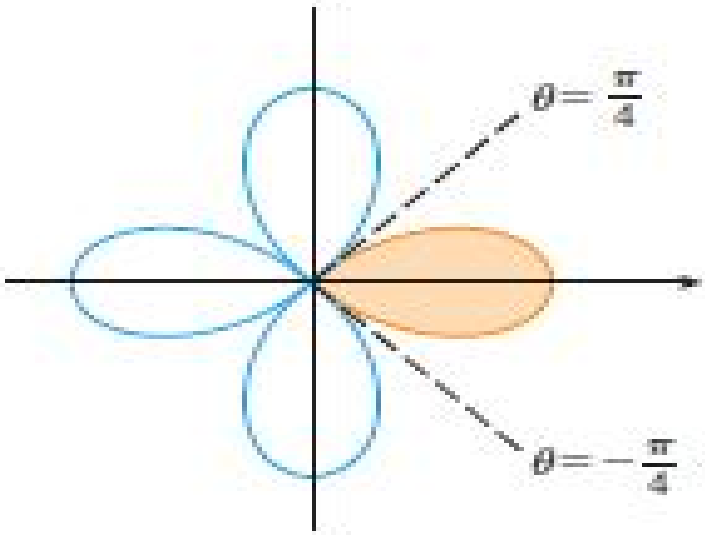


Figure 7:

From **Figure 7**, a loop is given by the region

$$D = \{(r, \theta) | \ -\pi/4 \leq \theta \leq \pi/4, \ 0 \leq r \leq \cos 2\theta\}$$

So the area is

$$A(D) = \iint_D \mathrm{d}A = \int_{-\pi/4}^{\pi/4} \int_0^{\cos 2\theta} r dr d\theta$$

$$= \int_{-\pi/4}^{\pi/4} \left[\frac{1}{2} r^2\right]\Big|_0^{\cos 2\theta} = \frac{1}{2} \int_{-\pi/4}^{\pi/4} \cos^2 2\theta d\theta = \frac{\pi}{8}.$$

Example 7. *Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$, above the xy-plane, and inside the cylinder $x^2 + y^2 = 2x$.*

Solution: The solid lies above the disk D whose boundary circle has equation $x^2 + y^2 = 2x$ (**Figure 8**).

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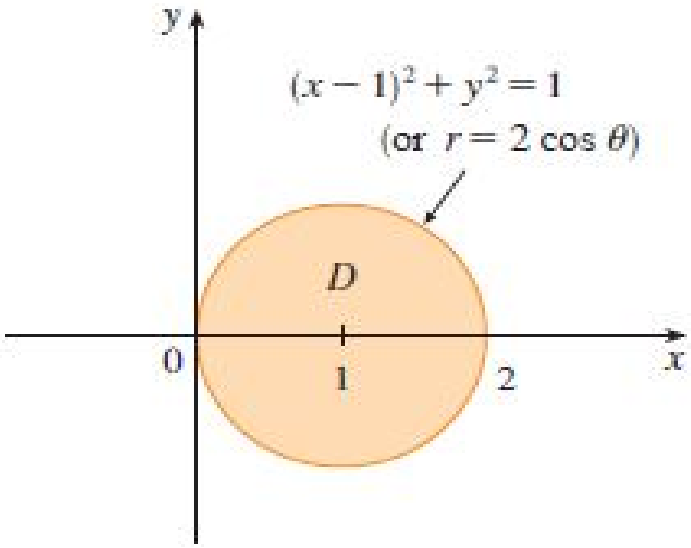


Figure 8:

In polar coordinates we have $x^2+y^2=r^2$ and $x=r\cos\theta$, so the boundary circle becomes $r^2=2r\cos\theta$, or $r=2\cos\theta$. Thus the disk D is given by

$$D=\{(r,\theta)|\ -\pi/2\le\theta\le\pi/2,\ 0\le r\le 2\cos\theta\}$$

Thus we have

$$V=\iint_D\left(x^2+y^2\right)\mathrm{d}A=\int_{-\pi/2}^{\pi/2}\int_0^{2\cos\theta}r^2 r\,dr\,d\theta=\int_{-\pi/2}^{\pi/2}\left[\frac{r^4}{4}\right]_0^{2\cos\theta}d\theta$$

$$=4\int_{-\pi/2}^{\pi/2}\cos^4\theta d\theta=8\int_0^{\pi/2}\cos^4\theta d\theta=8\int_0^{\pi/2}\left(\frac{1+\cos 2\theta}{2}\right)^2 d\theta$$

$$=2\int_0^{\pi/2}\left[1+2\cos 2\theta+\frac{1}{2}\Big(1+\cos 4\theta\Big)\right]d\theta=\frac{3\pi}{2}.$$

EXERCISES

(a) Use a double integral to nd the area of the region.

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(i) One loop of the rose $r = \cos 3\theta$.

(ii) The region enclosed by both of the cardioids $r = 1 + \cos\theta$ and $r = 1 - \cos\theta$

(iii) The region inside the circle $(x-1)^2 + y^2 = 1$ and outside the circle $x^2 + y^2 = 1$.

(b) Use polar coordinates to find the volume of the given solid.

(i) Under the cone $z = \sqrt{x^2 + y^2}$ and above the disk $x^2 + y^2 \leq 4$

(ii) Above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.

(iii) Inside both the cylinder $x^2 + y^2 = 4$ and the ellipsoid $4x^2 + 4y^2 + z^2 = 64$

(c) Use polar coordinates to combine the sum

$$\int_{\frac{1}{\sqrt{2}}}^{1}\int_{\sqrt{1-x^2}}^{x} xydydx + \int_{1}^{\sqrt{2}}\int_{0}^{x} xydydx + \int_{\sqrt{2}}^{2}\int_{0}^{\sqrt{4-x^2}} xydydx$$

into one double integral. Then evaluate the double integral.

0.1.4 Surface Area

The area of the surface with equation $z = f(x,y),\ (x,y) \in D$, where f_x and f_y are continuous is

$$A(S) = \iint_D \sqrt{\left[f_x(x,y)\right]^2 + \left[f_y(x,y)\right]^2 + 1}\,\mathrm{d}A.$$

If we use the alternative notation for partial derivatives, we can rewrite the formula as

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\,\mathrm{d}A.$$

Example 8. *Find the surface area of the part of the surface $z = x^2 + 2y$ that lies above the triangular region T in the xy-plane with vertices $(0,0)$, $(1,0)$, and $(1,1)$*

Solution: The region T is shown in **Figure 9** and is described by

$$T = \{(x,y)|\ 0 \leq x \leq 1,\ 0 \leq y \leq x\}.$$

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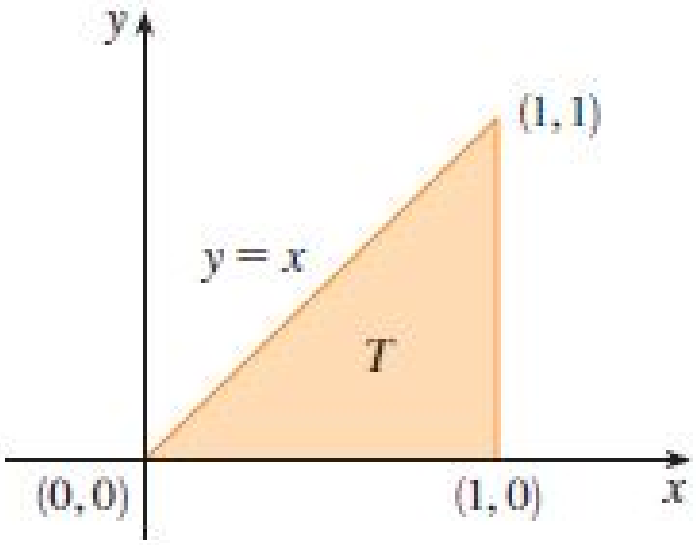


Figure 9:

With $f(x, y) = x^2 + 2y$, we have

$$A(S) = \iint_T \sqrt{(2x)^2 + (2)^2 + 1}\, \mathrm{d}A = \int_0^1 \int_0^x \sqrt{4x^2 + 5} dy dx$$

$$= \int_0^1 x\sqrt{4x^2 + 5} dx = \frac{1}{12}\Big(27 - 5\sqrt{5}\Big).$$

Example 9. *Find the area of the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 9$.*

Solution: The plane intersects the paraboloid in the circle $x^2 + y^2 = 9$, $z = 9$. Therefore the given surface lies above the disk D with center the origin and radius 3 **Figure 10.** Using the formula

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}\, \mathrm{d}A,$$

we have

$$A = \iint_D \sqrt{1 + (2x)^2 + (2y)^2}\, \mathrm{d}A = \iint_D \sqrt{1 + 4(x^2 + y^2)}\, \mathrm{d}A$$

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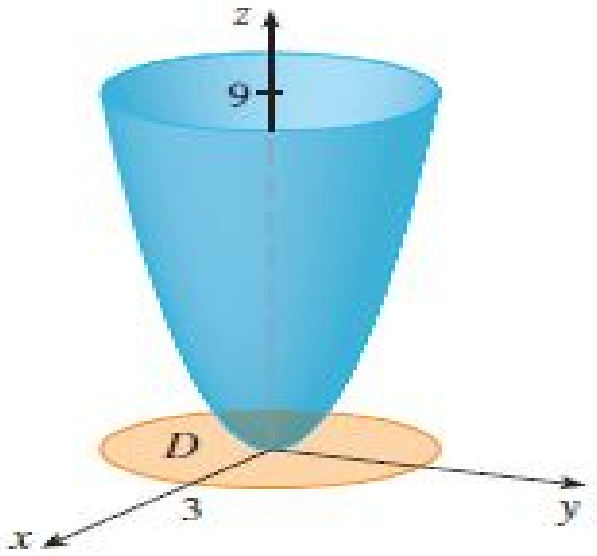


Figure 10:

Converting to polar coordinates, we obtain

$$A = \int_0^{2\pi} \int_0^3 \sqrt{1+4r^2} r dr d\theta = \int_0^{2\pi} d\theta \int_0^3 \frac{1}{8}\sqrt{1+4r^2}(8r)dr$$

$$= 2\pi\left(\frac{1}{8}\right)\frac{2}{3}\left(1+4r^2\right)^{\frac{3}{2}}\Big|_0^3 = \frac{\pi}{6}\left(37\sqrt{37}-1\right).$$

EXERCISES

(a) Find the area of the surface.

(i) The part of the plane $3x + 2y + z = 6$ that lies in the first octant.

(ii) The part of the surface $z = 1 + 3x + 2y^2$ that lies above the triangle with vertices $(0, 0)$, $(0, 1)$, *and* $(2, 1)$.

(iii) The part of the cylinder $y^2 + z^2 = 9$ that lies above the rectangle with vertices $(0, 0)$, $(4, 0)$, $(0, 2)$, *and* $(4, 2)$.

(iv) The part of the sphere $x^2 + y^2 + z^2 = 4z$ that lies inside the paraboloid $z = x^2 + y^2$.

(v) The part of the sphere $x^2 + y^2 + z^2 = a^2$ that lies within the cylinder $x^2 + y^2 = ax$ and above the xy-plane.

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(b) Show that the area of the part of the plane $z = ax + by + c$ that projects onto a region D in the xy-plane with area $A(D)$ is $\sqrt{a^2 + b^2 + 1}A(D)$.

0.2 Triple Integrals

Consider a simplest case where f is defined on a rectangular box:

$$B = \{(x, y, z) |\ a \le x \le b,\ c \le y \le d,\ r \le z \le s\}.$$

Divide the intervals $[a, b]$, $[c, d]$, *and* $[r, s]$ into l, m, *and* n subintervals of equal widths Δx, Δy, Δz respectively. The planes through the end points of these subintervals parallel to the coordinate planes divide the box B into l, m, n sub-boxes

$$B_{ijk} = [x_{i-1}, x_i] \times [y_{j-1}, x_j] \times [z_{k-1}, x_k].$$

Each sub-box has volume $\Delta V = \Delta x \Delta y \Delta z$. If we choose the sample point to be $(x^*_{ijk}, y^*_{ijk}, z^*_{ijk})$, we get an expression for the triple integral as follows:

$$V = \iiint_B f\big(x, y, z\big)\, \mathrm{d}V = \lim_{l,m,n\to\infty} \sum_{i=1}^{l} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk}) \Delta V.$$

The practical method for evaluating triple integrals is to express them as iterated integrals as follows.

Fubini's Theorem for Triple Integrals: If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f\big(x, y, z\big)\, \mathrm{d}V = \int_r^s \int_c^d \int_a^b f\big(x, y, z\big) dx dy dz.$$

Example 10. *Evaluate the triple integral*

$$\iiint_B xyz^2\, \mathrm{d}V,$$

where B is the rectangular box given by

$$B = \{(x, y, z) |\ 0 \le x \le 1,\ -1 \le y \le 2,\ 0 \le z \le 3\}.$$

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Solution:

$$\iiint_B xyz^2 \, \mathrm{d}V = \int_0^3 \int_{-1}^2 \int_0^1 xyz^2 \, \mathrm{d}x\mathrm{d}y\mathrm{d}z$$

$$\int_0^3 \int_{-1}^2 \left[\frac{x^2yz^2}{2}\right]\Bigg|_{x=0}^{x=1} dydz = \int_0^3 \int_{-1}^2 \frac{yz^2}{2} dydz = \int_0^3 \left[\frac{y^2z^2}{4}\right]\Bigg|_{y=-1}^{y=2} dz$$

$$\int_0^3 \frac{3z^2}{4} dz = \frac{27}{4}.$$

Example 11. *Evaluate*

$$\iiint_E z \, \mathrm{d}V,$$

where E is the solid tetrahedron bounded by the four planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 1$.

Solution: The solid region is shown in **Figure** 11 and one of its projection D onto the xy-plane in **Figure** 12.

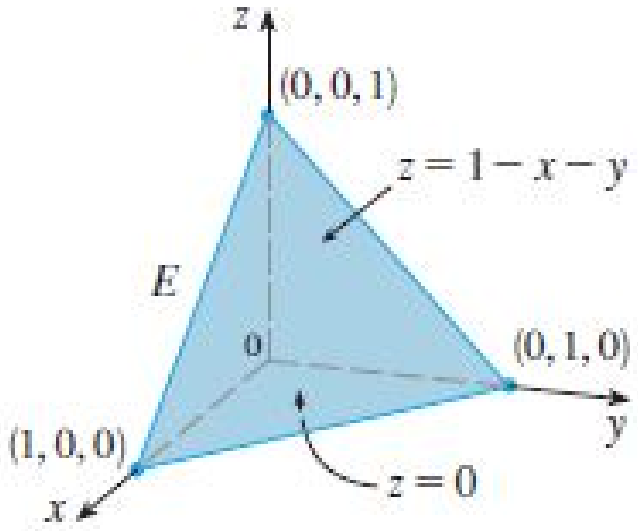


Figure 11:

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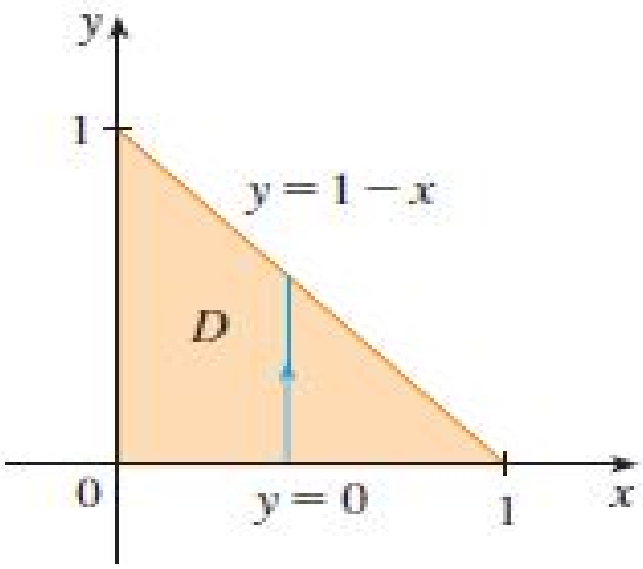


Figure 12:

The lower boundary of the tetrahedron is the plane $z = 0$ and the upper boundary is the plane $x + y + z = 1$, $(or\ \ z = 1 - x - y)$, so we have

$$E = \{(x, y, z)|\ 0 \le x \le 1,\ \ 0 \le y \le 1 - x,\ \ 0 \le z \le 1 - x - y\}, \tag{6}$$

Thus we have

$$\iiint_E z\,\mathrm{d}V = \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z dz dy dx = \frac{1}{24}.$$

EXERCISES

(a) Evaluate the triple integral.

(i)

$$\iiint_E y\,\mathrm{d}V,$$

where

$$E = \{(x, y, z)|\ 0 \le x \le 3,\ \ 0 \le y \le x\ \ x - y \le z \le x + y\},$$

(ii)

$$\iiint_E \frac{z}{x^2 + z^2}\,\mathrm{d}V,$$

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where

$$E = \{(x, y, z) |\ 1 \le y \le 4,\ \ y \le z \le 4,\ \ 0 \le x \le z\},$$

(iii)

$$\iiint_E 6xy \, \mathrm{d}V,$$

where E lies under the plane $z = 1 + x + y$ and above the region in the xy-plane bounded by the curves $y = \sqrt{x},\ \ y = 0,\ \ and\ x = 1$.

(b) Use a triple integral to find the volume of the given solid.

(i) The tetrahedron enclosed by the coordinate planes and the plane $2x + y + z = 4$.

(ii) The solid enclosed by the cylinder $x^2 + z^2 = 4$ and the planes $y = -1$ and $y + z = 4$.

(iii) The solid enclosed by the paraboloids $y = x^2 + z^2$ and $y = 8 - x^2 - z^2$.

0.2.1 Triple Integrals in Cylindrical Coordinates

Figure 13 enables us to recall the connection between polar and Cartesian coordinates.

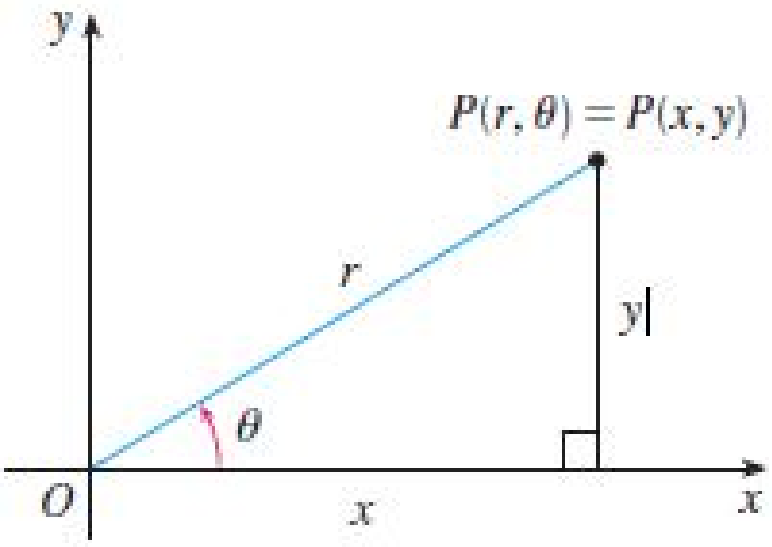


Figure 13:

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If the point P has Cartesian coordinates (x, y) and polar coordinates (r, θ), then from the figure,

$$x = r\cos\theta, \quad y = r\sin\theta,$$

$$r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}.$$

In the cylindrical coordinate system, a point P in three-dimensional space is represented by the ordered triple (r, θ, z), where r and θ are polar coordinates of the projection of P onto the xy-plane and z is the directed distance from the xy-plane to P (**Figure 14**).

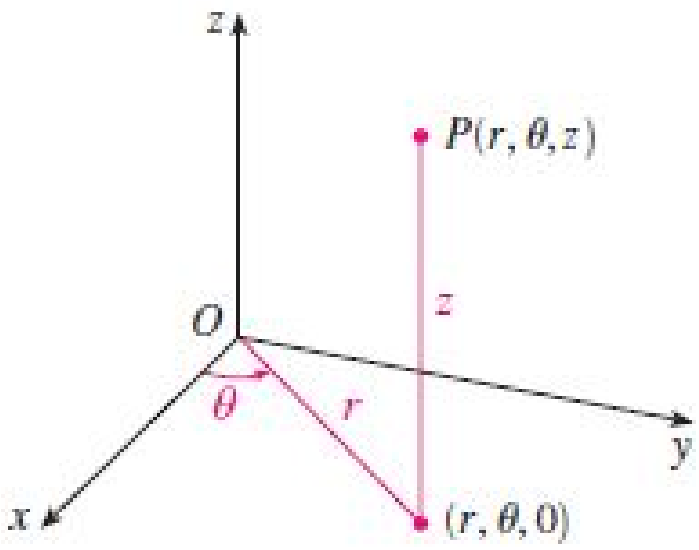


Figure 14:

To convert from cylindrical to rectangular coordinates, we use the equations

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z$$

whereas to convert from rectangular to cylindrical coordinates, we use

$$r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}, \quad z = z.$$

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0.2.2 Evaluating Triple Integrals with Cylindrical Coordinates

Suppose that E is a type 1 region whose projection D onto the xy-plane is conveniently described in polar coordinates (**Figure 15**).

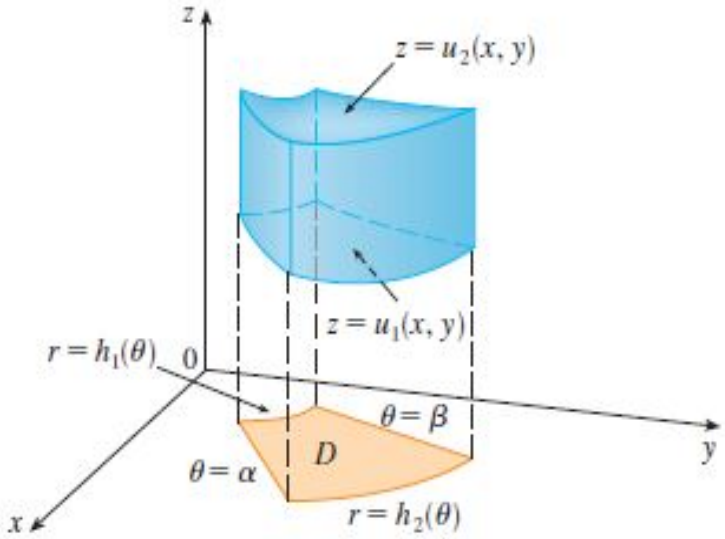


Figure 15:

In particular, suppose that f is continuous and

$$E = \{(x, y, z) |\ (x, y) \in D,\ \ u_1(x, y) \leq z \leq u_2(x, y)\}, \tag{7}$$

where D is given in polar coordinates by

$$D = \{(r, \theta) |\ \ \alpha \leq \theta \leq \beta\ \ \ h_1(\theta) \leq r \leq h_2(\theta)\}.$$

Thus in form of cylindrical coordinates, the triple integral is

$$\iiint_E f(x, y, z)\, \mathrm{d}V = \int_\alpha^\beta \int_{h_1(\theta)}^{h_2(\theta)} \int_{u_1(r\cos\theta, r\sin\theta)}^{u_2(r\cos\theta, r\sin\theta)} f(r\cos\theta, r\sin\theta, z) r dz dr d\theta.$$

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Example 12. *Evaluate*

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}\left(x^2+y^2\right)dzdydx$$

Solution: This iterated integral is a triple integral over the solid region

$$E=\{(x,y,z)|\ \ -2\le x\le 2,\ \ -\sqrt{4-x^2}\le y\le\sqrt{4-x^2},\ \ -\sqrt{x^2+y^2}\le z\le 2\},$$

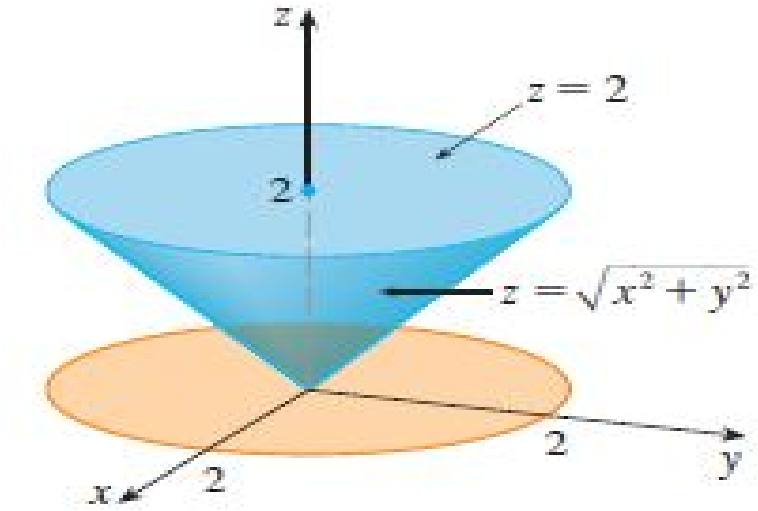


Figure 16:

and the projection of E onto the xy-plane is the disk $x^2+y^2\le 4$. The lower surface of E is the cone $z=\sqrt{x^2+y^2}$ and its upper surface is the plane $z=2$ (**Figure 16**). This region has a much simpler description in cylindrical coordinates:

$$E=\{(x,y,z)|\ \ 0\le x\le 2\pi,\ \ 0\le r\le 2,\ \ r\le z\le 2\},$$

Therefore we have

$$\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)dzdydx=\iiint_E(x^2+y^2)\,\mathrm{d}V=$$

$$=\int_0^{2\pi}\int_0^2\int_r^2 r^2 r dzdrd\theta=\frac{16}{5}\pi.$$

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EXERCISES

(a) Use cylindrical coordinates.

(i) Evaluate

$$\iiint_E (x^2 + y^2)\,\mathrm{d}V,$$

where E is the region that lies inside the cylinder $x^2 + y^2 = 16$ and between the planes $z = -5$ and $z = 4$.

(ii) Evaluate

$$\iiint_E x^2\,\mathrm{d}V,$$

where E is the solid that lies within the cylinder $x^2 + y^2 = 1$, and above the plane $z = 0$, and below the cone $z^2 = 4x^2 + 4y^2$.

(iii) Find the volume of the solid that lies within both the cylinder $x^2 + y^2 = 1$ and the sphere $x^2 + y^2 + z^2 = 4$.

(iv) Find the volume of the solid that lies between the paraboloid $z = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 2$.

0.2.3 Triple Integrals in Spherical Coordinates

Spherical Coordinates: The **spherical coordinates** (ρ, θ, ϕ) of a point P in space are shown in **Figure 17**.

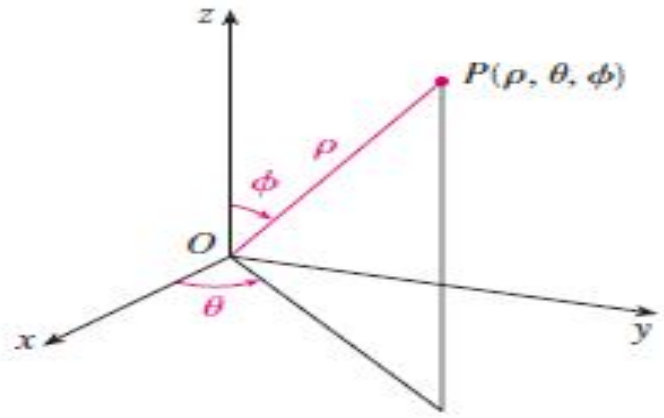


Figure 17:

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where $\rho = |OP|$ is the distance from the origin to P, θ is the same angle as in cylindrical coordinates, and ϕ s the angle between the positive z-axis and the line segment OP.
Note that $\rho \geq 0 \qquad 0 \leq \phi \leq \pi$.
The spherical coordinate system is especially useful in problems where there is symmetry about a point, and the origin is placed at this point. The relationship between rectangular and spherical coordinates can be seen in **Figure 18**

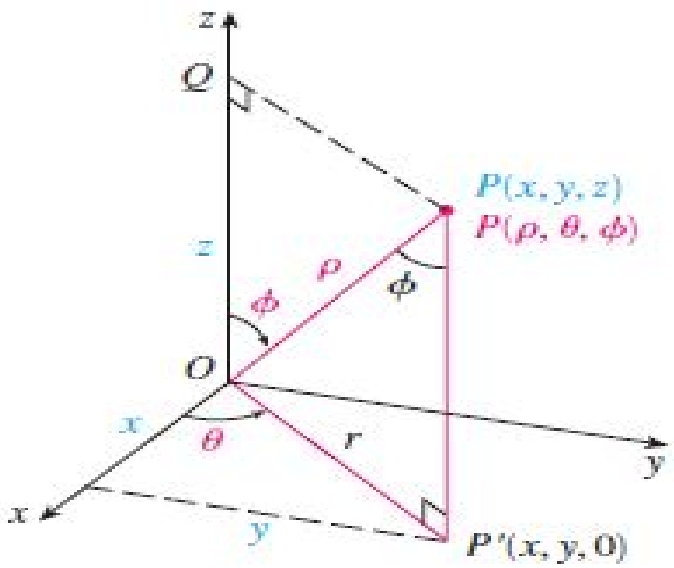


Figure 18:

From triangles OPQ and OPP' we have

$$z = \rho\cos\phi \qquad r = \rho\sin\phi.$$

But $x = r\cos\theta$ and $y = r\sin\theta$, so to convert from spherical to rectangular coordinates, we use the equations

$$x = \rho\sin\phi\cos\theta, \qquad x = \rho\sin\phi\sin\theta, \qquad z = \rho\cos\phi.$$

Also, the distance formula shows that

$$\rho^2 = x^2 + y^2 + z^2.$$

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Evaluating Triple Integrals with Spherical Coordinates

In the spherical coordinate system the counterpart of a rectangular box is a **spherical wedge**

$$E = \{(\rho, \theta, \phi) |\ a \le \rho \le b,\ \ \alpha \le \theta \le \beta,\ \ c \le \phi \le d\},$$

We use this equation in converting from rectangular to spherical coordinates.
The volume element in spherical coordinates

$$dV = \rho^2 \sin\phi d\rho d\theta d\phi.$$

Consequently, we have the following formula for triple integration in **spherical coordinates**:

$$\iiint_E f(x, y, z)\, \mathrm{d}V = \int_c^d \int_\alpha^\beta \int_a^b f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2 \sin\phi d\rho d\theta d\phi,$$

where E is a spherical wedge given by

$$E = \{(\rho, \theta, \phi) |\ a \le \rho \le b,\ \ \alpha \le \theta \le \beta,\ \ c \le \phi \le d\},$$

Example 13. *Evaluate*

$$\iiint_B e^{(x^2+y^2+z^2)^{3/2}}\, \mathrm{d}V,$$

where B is the unit ball:

$$E = \{(x, y, z) |\ x^2 + y^2 + z^2 \le 1\},$$

Solution: Since the boundary of B is a sphere, we use spherical coordinates:

$$E = \{(\rho, \theta, \phi) |\quad 0 \le \rho \le 1, \quad 0 \le \theta \le 2\pi, \quad 0 \le \phi \le \pi\}.$$

In addition, spherical coordinates are appropriate because

$$x^2 + y^2 + z^2 = \rho^2$$

Thus we have

$$\iiint_B e^{(x^2+y^2+z^2)^{3/2}}\, \mathrm{d}V = \int_0^\pi \int_0^{2\pi} \int_0^1 e^{(\rho^2)^{3/2}} \rho^2 \sin\phi d\rho d\theta d\phi$$

$$= \int_0^\pi \sin\phi d\phi \int_0^{2\pi} d\theta \int_0^1 \rho^2 e^{\rho^3} d\rho$$

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$$= \left[-\cos\phi \right]_0^{\pi} (2\pi) \left[\frac{1}{3} e^{\rho^3} \right]_0^1 = \frac{4}{3}\pi(e-1).$$

EXERCISES

(a) Use spherical coordinates.

(i) Evaluate

$$\iiint_E (x^2+y^2+z^2)^2 \, \mathrm{d}V,$$

where B is the ball with center the origin and radius 5.

(ii) Evaluate

$$\iiint_H (9-x^2-y^2)^2 \, \mathrm{d}V,$$

where H is the solid hemisphere $x^2+y^2+z^2 \le 9, \quad z \ge 0$.

(iii) Evaluate

$$\iiint_E x e^{x^2+y^2+z^2} \, \mathrm{d}V,$$

where E is the portion of the unit ball $x^2+y^2+z^2 \le 1$ that lies in the first octant.

(iv) Find the volume of the solid that lies within the sphere $x^2+y^2+z^2 \le 4$, above the xy-plane, and below the cone $z = \sqrt{x^2+y^2}$.

(b) Show that

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \sqrt{x^2+y^2+z^2}\, e^{-(x^2+y^2+z^2)} dx dy dz = 2\pi.$$

(The improper triple integral is defined as the limit of a triple integral over a solid sphere as the radius of the sphere increases indefinitely.)

0.3 Change of Variables in Multiple Integrals

We consider a change of variables that is given by a **transformation** T from the uv-plane to the xy-plane:

$$T(u,v) = (x,y),$$

where x and y are related to u and v by the equations

$$x = g(u,v), \quad y = h(u,v)$$

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or, as we sometimes write,

$$x = x(u, v), \quad y = y(u, v).$$

We usually assume that T is a **$\mathbf{C^1}$ transformation**, which means that g and h have continuous first-order partial derivatives.

Definition 1. *The* **Jacobian** *of the transformation T given by $x = g(u, v)$ and $y = h(u, v)$ is*

$$\frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \tag{8}$$

Change of Variables in a Double Integral:

Theorem 1. *Suppose that T is a C^1 transformation whose Jacobian is nonzero and that maps a region S in the uv-plane onto a region R in the xy-plane. Suppose that f is continuous in R and that R and S are type I or type II plane regions. Suppose also that T is a one-to-one, except perhaps on the boundary of S. Then*

$$\iint_R f(x, y)\,\mathrm{d}A = \iint_S f\Big((u, v), y(u, v)\Big)\left|\frac{\partial(x, y)}{\partial(u, v)}\right| \mathrm{d}u\,\mathrm{d}v.$$

The theorem says that we change from an integral in x and y to an integral in u and v by expressing x and y in terms of u and v and writing

$$dA = \left|\frac{\partial(x, y)}{\partial(u, v)}\right| \mathrm{d}u\,\mathrm{d}v.$$

Example 14. *Show that when changing to polar coordinates we have $dA = rdrd\theta$.*

Solution: The transformation here is the standard conversion formulae:

$$x = r\cos\theta, \quad y = r\sin\theta$$

The Jacobian for this transformation is

$$\begin{aligned} \frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} &= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} \\ &= r\cos^2\theta - (-r\sin^2\theta) \\ &= r(\cos^2\theta + \sin^2\theta) = r. \end{aligned}$$

Thus we get

$$dA = \left|\frac{\partial(x, y)}{\partial(r, \theta)}\right| \mathrm{d}r\,\mathrm{d}\theta = |r|\mathrm{d}r\,\mathrm{d}\theta = r\mathrm{d}r\,\mathrm{d}\theta.$$

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Example 15. *Evaluate*

$$\iint_R (x+y)\,\mathrm{d}A,$$

where R is the trapezoidal region with vertices given by $(0,0)$, $(5,0)$, $(5/2,5/2)$, and $(5/2,-5/2)$ using the transformation $x=2u+3v$ and $y=2u-3v$.

Solution: In **Figure 19**,

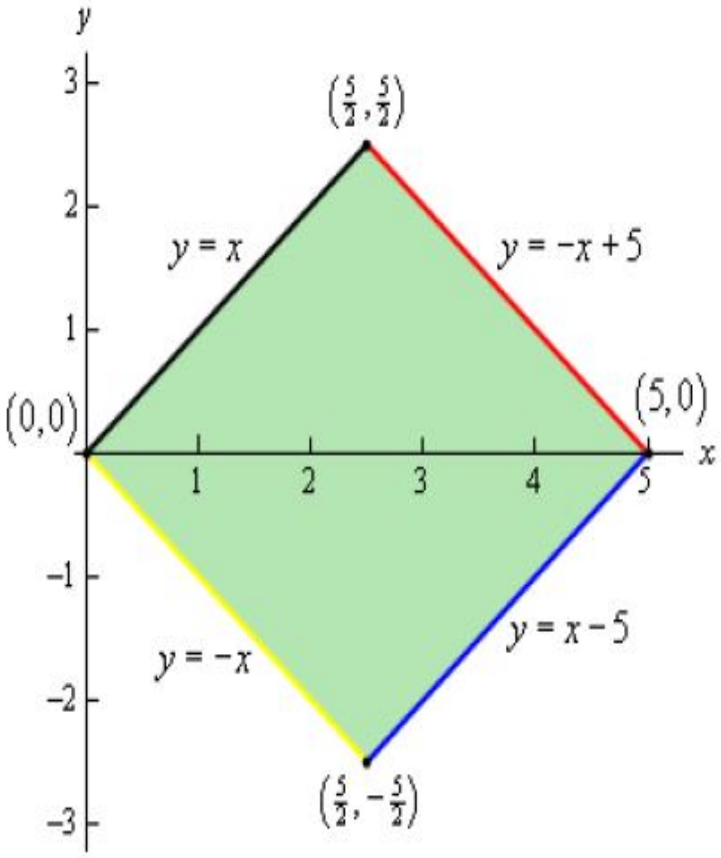


Figure 19:

with $y=x$, we have

$$2u-3v=2u+3v$$
$$6v=0 \implies v=0.$$

Transforming y=-x, we obtain

$$2u-3v=-(2u+3v)$$

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$$4u = 0 \implies u = 0.$$

We next transform $y = -x + 5$, to obtain

$$2u - 3v = -(2u + 3v) + 5$$
$$4u = 5 \implies u = \frac{5}{4}.$$

Also transforming $y = x - 5$, we get

$$2u - 3v = 2u + 3v - 5$$
$$-6v = -5 \implies v = \frac{5}{6}.$$

The region S is then a rectangle whose sides are given by $u = 0$, $v = 0$, $u = \frac{5}{4}$ and $v = \frac{5}{6}$ and so the ranges u and v are

$$0 \le u \le \frac{5}{4} \qquad 0 \le v \le \frac{5}{6}.$$

The Jacobian is obtained as

$$\frac{\partial(x, y)}{\partial(u, v)} = -12.$$

The integral is then

$$\iint_R (x + y)\,\mathrm{d}A = \int_0^{\frac{5}{6}} \int_0^{\frac{5}{4}} \Big((2u + 3v) + (2u - 3v) \Big) \Big| - 12 \Big| \,\mathrm{d}u\,\mathrm{d}v$$
$$= \int_0^{\frac{5}{6}} \int_0^{\frac{5}{4}} 48u\,\mathrm{d}u\,\mathrm{d}v = \frac{125}{4}.$$

Triple Integrals

There is a similar change of variables formula for triple integrals. Let T be a transformation that maps a region S in uvw-space onto a region R in xyz-space by means of the equations

$$x = g(u, v, w) \qquad y = h(u, v, w) \qquad z = k(u, v, w).$$

The **Jacobian** of T is the following 3×3 determinant:

$$\frac{\partial(x, y, z)}{\partial(u, v, w)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Thus we have the following formula for triple integrals:

$$\iiint_R f(x, y, z)\,\mathrm{d}V = \iiint_S f\Big((u, v, w), y(u, v, w), z(u, v, w) \Big) \left| \frac{\partial(x, y, z)}{\partial(u, v, w)} \right| \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w.$$

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EXERCISES

(a) Use the given transformation to evaluate the integral.

(i)
$$\iint_R (x - 3y)\,\mathrm{d}A,$$
where R is a triangular region with vertices $(0,0)$, $(2,1)$, and $(1,2)$; $x = 2u + v,\ \ y = u + 2v$.

(ii)
$$\iint_R (4x + 8y)\,\mathrm{d}A,$$
where R is the parallelogram with vertices $(-1,3)$, $(1,-3)$, $(3,-1)$, and $(1,5)$; $x = \frac{1}{4}(u + v)$, $y = \frac{1}{4}(v - 3u)$.

(iii)
$$\iint_R x^2\,\mathrm{d}A,$$
where R is the region bounded by the ellipse $9x^2 + 4y^2 = 36;\ x = 2u,\ \ y = 3v$.

(b) Evaluate the integral by making an appropriate change of variables.

(i)
$$\iint_R \frac{x - 2y}{3x - y}\,\mathrm{d}A,$$
where R is a parallelogram enclosed by the lines $x - 2y = 0,\ x - 2y = 4,\ 3x - y = 1$ *and* $3x - y = 8$.

(ii)
$$\iint_R (x + y\,)e^{x^2 - y^2}\,\mathrm{d}A,$$
where R is a rectangle enclosed by the lines $x - y = 0,\ x - y = 2,\ x + y = 0$ *and* $x + y = 3$.

(iii)
$$\iint_R \cos\left(\frac{y - x}{y + x}\right)\mathrm{d}A,$$
where R is the trapezoidal region with vertices $(1,0),\ (2,0),\ (0,2)$ *and* $(0,1)$.

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(c) Use the transformation $u = x - y,\ v = x + y$ to evaluate

$$\iint_R \left(\frac{x-y}{x+y}\right) \mathrm{d}A,$$

where R is the square with vertices $(0,2),\ (1,1),\ (2,2)$ *and* $(1,3)$.

0.4 Line Integrals

If f is defined on a smooth curve C given by the parametric equations

$$x = x(t), \qquad y = y(t), \qquad a \le t \le b,$$

then the **line integral of f along C** is

$$\int_C f(x,y)ds = \int_a^b f\big(x(t),y(t)\big)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt \tag{9}$$

Example 16. *Evaluate* $\int_C (2+x^2y)ds$, *where* C *is the upper half of the unit circle* $x^2+y^2 = 1$.

Solution: Recall that the unit circle can be parametrized by means of the equations

$$x = \cos t, \qquad y = \sin t$$

and the upper half of the circle is described by the parameter interval $0 \le t \le \pi$. (See Figure 20)

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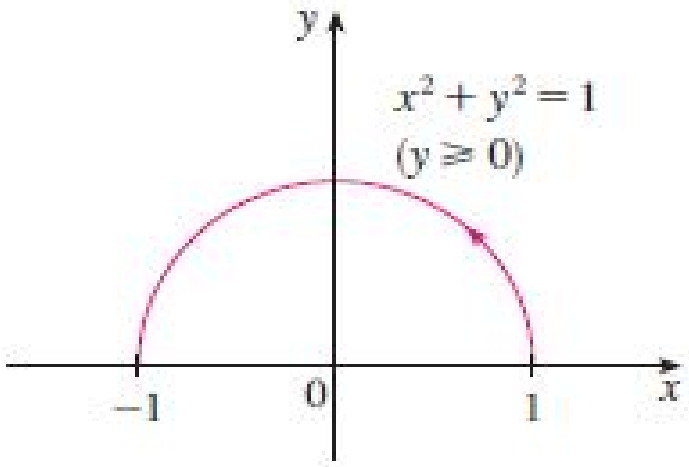


Figure 20:

Therefore we have

$$\int_C (2 + x^2 y)ds = \int_0^\pi \left(2 + \cos^2 t \sin t\right) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^\pi \left(2 + \cos^2 t \sin t\right) \sqrt{\sin^2 t + \cos^2 t}\, dt = \int_0^\pi \left(2 + \cos^2 t \sin t\right) = 2\pi + \tfrac{2}{3}.$$

Suppose that C is a **piecewise-smooth curve;** that is, C is a union of a finite number of smooth curves $C_1, C_2, \cdots, C_n$, where as illustrated in **Figure** 21,

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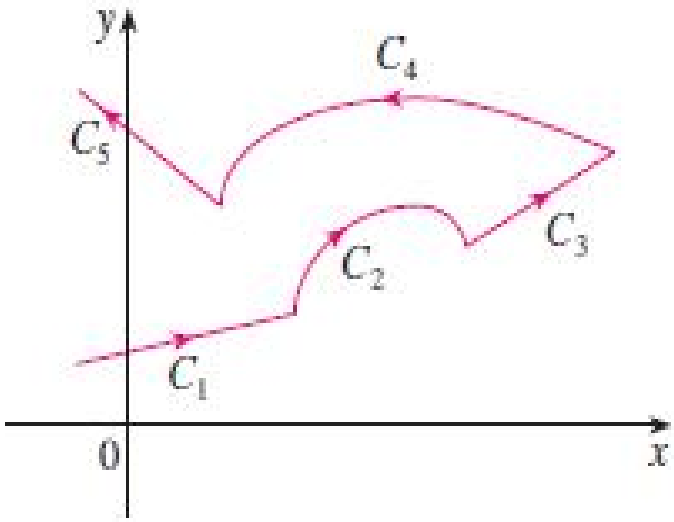


Figure 21:

the initial point of C_{i+1} is the terminal point of C_i. Then we define the integral of f along C as the sum of the integrals of f along each of the smooth pieces of C :

$$\int_C f(x,y)ds = \int_{C_1} f(x,y)ds + \int_{C_2} f(x,y)ds + \cdots + \int_{C_n} f(x,y)ds.$$

Example 17. *Evaluate* $\int_C 2xds$, *where* C *consists of the arc* C_1 *of the parabola* $y = x^2$ *from* $(0,0)$ *to* $(1,1)$ *followed by the vertical line segment* C_2 *from* $(1,1)$ *to* $(1,2)$.

Solution: The curve C is shown in Figure 22.

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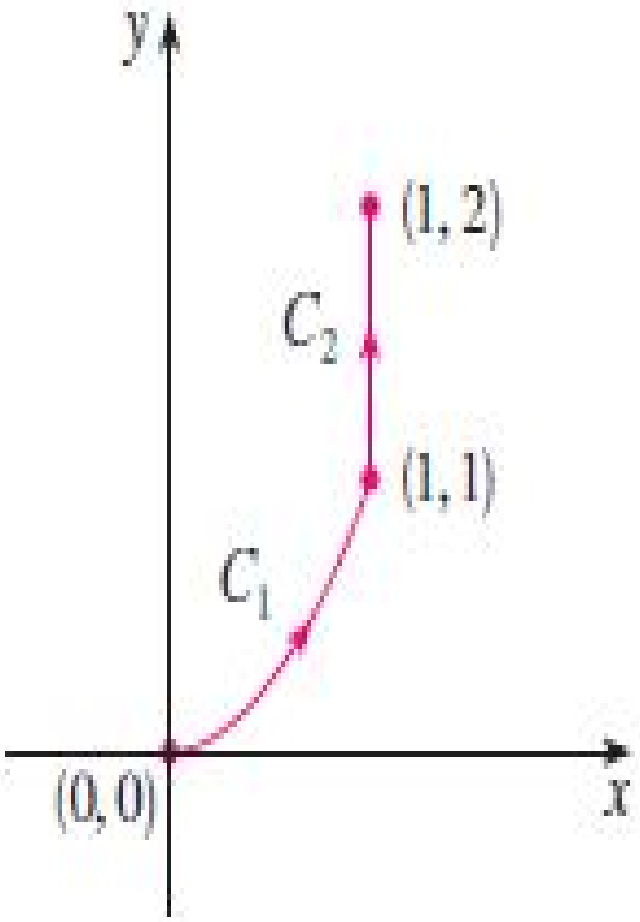


Figure 22:

C_1 is the graph of a function of x, so we can choose x as the parameter and the equations for C_1 become

$$x = x, \quad y = x^2, \quad 0 \le x \le 1.$$

Therefore

$$\int_{C_1} 2x\,ds = \int_0^1 2x\sqrt{\left(\frac{dx}{dx}\right)^2 + \left(\frac{dy}{dx}\right)^2}\,dx = \int_0^1 2x\sqrt{1+4x^2}\,dx$$

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$$= \frac{1}{4}\left(\frac{2}{3}\right)(1+4x^2)^{3/2}\Big|_0^1 = \frac{5\sqrt{5}-1}{6}.$$

On C_2, we choose y as the parameter, so the equations of C_2 are

$$x = 1, \quad y = y, \quad 1 \le y \le 2$$

and

$$\int_{C_2} 2x\,ds = \int_1^2 2(1)\sqrt{\left(\frac{dx}{dy}\right)^2 + \left(\frac{dy}{dy}\right)^2}\,dy = \int_1^2 2\,dy = 2.$$

Thus

$$\int_C 2x\,ds = \int_{C_1} 2x\,ds + \int_{C_2} 2x\,ds = \frac{5\sqrt{5}-1}{6} + 2.$$

When we are setting up a line integral, sometimes the most difficult thing is to think of a parametric representation for a curve whose geometric description is given. In particular, we often need to parametrize a line segment, so it's useful to remember that a vector representation of the line segment that starts at $\mathbf{r}_0$ and end at $\mathbf{r}_1$ is given by

$$\mathbf{r}(t) = (1-t)\mathbf{r}_0 + t\mathbf{r}_1. \tag{10}$$

Example 18. *Evaluate $\int_C y^2\,dx + x\,dy$, where (a) $C = C_1$ is the line segment from $(-5,-3)$ to $(0,2)$ and (b) $C = C_2$ s the arc of the parabola $x = 4 - y^2$ from $(-5,-3)$ to $(0,2)$. (See Figure 23.)*

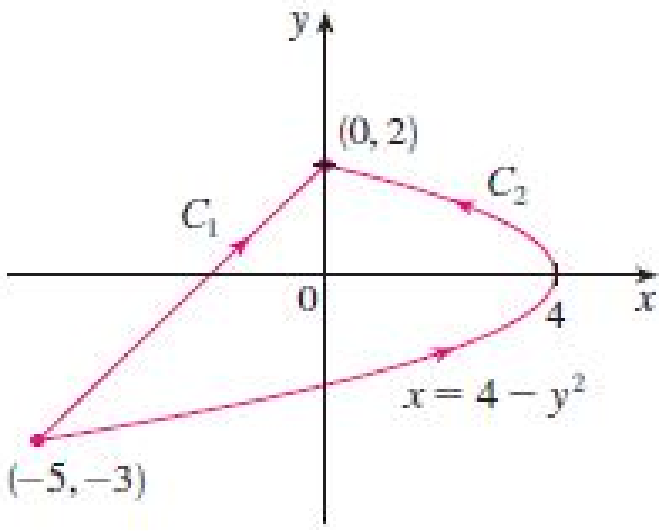


Figure 23:

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Solution: (a) A parametric representation for the line segment is

$$x = 5t - 5, \quad y = 5t - 3, \quad 0 \le t \le 1.$$

Here $\mathbf{r}_0 = \langle -5, -3 \rangle$, $\mathbf{r}_1 = \langle 0, 2 \rangle$. Then $dx = 5\,dt$, $dy = 5\,dt$ and thus we have

$$\int_{C_1} y^2\,dx + x\,dy = \int_0^1 (5t-3)^2(5\,dt) + (5t-5)(5\,dt)$$

$$= 5\int_0^1 (25t^2 - 25t + 4)\,dt = -\frac{5}{6}.$$

(b) Since the parabola is given as a function of y, let's take y as the parameter and write C_2 as

$$x = 4 - y^2, \quad y = y, \quad -3 \le y \le 2.$$

Then $dx = -2y\,dy$ and thus we have

$$\int_{C_2} y^2\,dx + x\,dy = \int_{-3}^2 y^2(-2y)\,dy + (4-y^2)\,dy = 40\frac{5}{6}.$$

0.4.1 Line Integrals in Space

Suppose that C is a smooth space curve given by the parametric equations

$$x = x(t), \qquad y = y(t), \qquad z = z(t), \qquad a \le t \le b,$$

or by a vector equation $\mathbf{r}(t)=x(t)\mathbf{i}+y(t)\mathbf{j}+z(t)\mathbf{k}$. If f is a function of three variables that is continuous on some region containing C, hen we define the **line integral of f along C** (with respect to arc length) in a manner similar to that for plane curves.
Therefore we have

$$\int_C f(x,y,z)ds = \int_a^b f\Big(x(t),y(t),z(t)\Big)\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\,dt \qquad (11)$$

Example 19. *Evaluate $\int_C ydx + zdy + xdz$, where C consists of the line segment C_1 from $(2,0,0)$ to $(3,4,5)$, followed by the vertical line segment C_2 from $(3,4,5)$ to $(3,4,0)$.*

Solution: The curve C is shown in Figure 24

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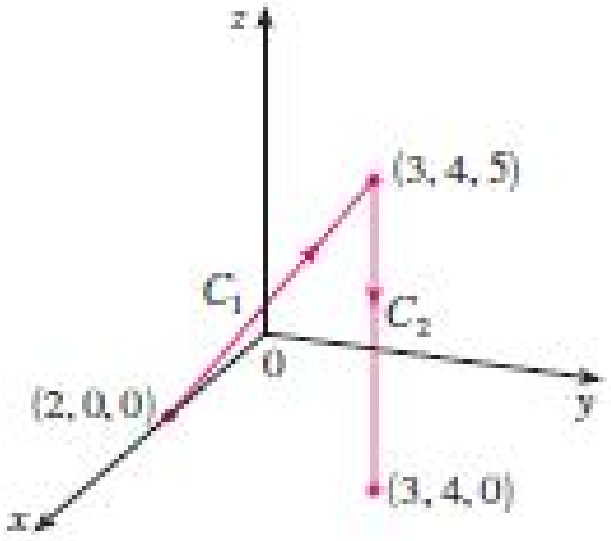


Figure 24:

Using Equation 10, we write

$$\mathbf{r}(t) = (1-t)\langle 2,0,0\rangle + t\langle 3,4,5\rangle = \langle 2+t, 4t, 5t\rangle$$

or, in parametric form, as

$$x = 2+t, \quad y = 4t, \quad z = 5t, \quad 0 \le t \le 1.$$

Thus

$$\int_C y\,dx + z\,dy + x\,dz = \int_0^1 (4t)\,dt + (5t)4\,dt + (2+t)5\,dt$$
$$= \int_0^1 (10+29t)\,dt = 24.5$$

Likewise, C_2 can be written in the form

$$\mathbf{r}(t) = (1-t)\langle 3,4,5\rangle + t\langle 3,4,0\rangle = \langle 3,4,5-5t\rangle$$

or

$$x = 3, \quad y = 4, \quad z = 5-5t, \quad 0 \le t \le 1.$$

Then $dx = 0 = dy$, so

$$\int_{C_2} ydx + zdy + xdz = \int_0^1 3(-5)\,dt = -15.$$

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Adding the values of these integrals, we obtain

$$\int_C ydx + zdy + xdz = 24.5 - 15 = 9.5$$

0.4.2 Line Integrals in Vector Fields

Definition 2. *Let* $\mathbf{F}$ *be a continuous vector field defined on a smooth curve* C *given by a vector function* $\mathbf{r}(t)$*,* $a \leq t \leq b$*. Then the* **line integral of F along C** *is*

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(\mathbf{t})) \cdot \mathbf{r}'(t)dt = \int_C \mathbf{F} \cdot \mathbf{T}ds.$$

Example 20. *Find the work done by the force field* $\mathbf{F}(x, y) = x^2\mathbf{i} - xy\mathbf{j}$ *in moving a particle along the quarter-circle* $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$*,* $0 \leq t \leq \frac{\pi}{2}$.

Solution: Since $x = \cos t$ and $y = \sin t$, we have

$$\mathbf{F}(\mathbf{r}(t)) = \cos^2 t\mathbf{i} - \cos t \sin t\mathbf{j}$$

and

$$\mathbf{r}'(t) = -\sin t\mathbf{i} + \cos t\mathbf{j}.$$

Therefore the work done is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{\pi/2} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)dt = \int_0^{\pi/2} \Big(-2\cos^2 t \sin t\Big)dt = -\frac{2}{3}.$$

Suppose the vector field $\mathbf{F}$ on $\mathbb{R}^3$ is given in component form by the equation $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C Pdx + Qdy + Rdz, \text{ where } \vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}.$$

EXERCISES

(a) Evaluate the line integral, where C is the given curve.

(i) $\int_C xy^4 ds$, C is the right half of the circle $x^2 + y^2 = 16$.

(ii) $\int_C (x + 2y)dx + x^2 dy$, C consists of line segments from $(0, 0)$ to $(2, 1)$ and from $(2, 1)$ to $(3, 0)$

(iii) $\int_C xe^{yz} ds$, C is the line segment from $(0, 0, 0)$ to $(1, 2, 3)$.

(b) Find the work done by the force field $\mathbf{F}(x, y) = x\mathbf{i} + (y + 2)\mathbf{j}$ in moving an object along an arch of the cycloid $\mathbf{r}(t) = (t - \sin t)\mathbf{i} + (1 - \cos t)\mathbf{j}$, $0 \leq t \leq 2\pi$.

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0.5 Fundamental Theorem for Line Integrals

Theorem 2. *Let C be a smooth curve given by the vector function $\mathbf{r}(t)$, $a \leq t \leq b$. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C. Then*

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

If f is a function of two variables and C is a plane curve with initial point $A(x_1, y_1)$ and terminal point $B(x_2, y_2)$ as in Figure 6, then Theorem 2 becomes

$$\int_C \nabla f \cdot d\mathbf{r} = f(x_2, y_2) - f(x_1, y_1).$$

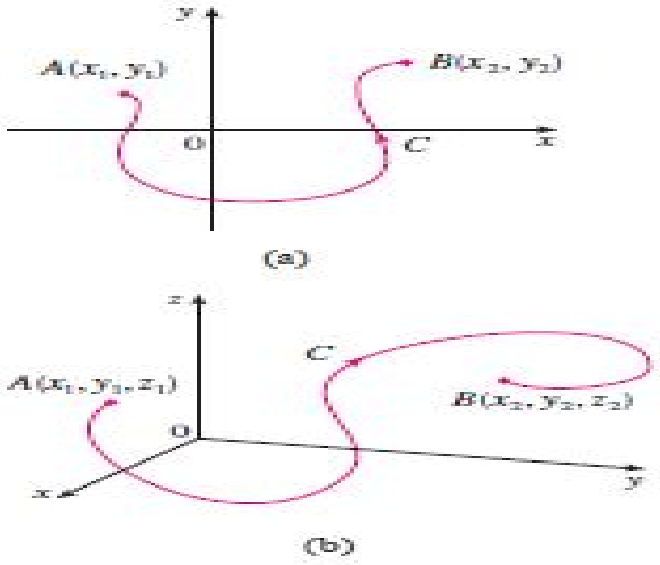


Figure 25:

If f is a function of three variables and C is a space curve joining the point $A(x_1, y_1, z_1)$ to the point $B(x_2, y_2, z_2)$, then we have

$$\int_C \nabla f \cdot d\mathbf{r} = f(x_2, y_2, z_2) - f(x_1, y_1, z_1).$$

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Proof. Using Definition 1, we have

$$\int_C \nabla f \cdot d\mathbf{r} = \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$$

$$= \int_a^b \left(\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} \right) dt$$

$$= \int_a^b \frac{d}{dt} f(\mathbf{r}(t))\, dt$$

$$= f(\mathbf{r}(b)) - f(\mathbf{r}(a)).$$

□

Example 21. *Evaluate* $\int_C \nabla f \cdot d\mathbf{r}$, *where* $f(x, y, z) = \cos(\pi x) + \sin(\pi y) - xyz$ *and* C *is any path that starts at* $(1, 1/2, 2)$ *and ends at* $(2, 1, -1)$.

Solution: Let $\mathbf{r}(t)$, $a \le t \le b$ be any path that starts at $(1, 1/2, 2)$ and ends at $(2, 1, -1)$. Then

$$\mathbf{r}(a) = \Big\langle 1, 1/2, 2 \Big\rangle, \quad \mathbf{r}(b) = \Big\langle 2, 1, -1 \Big\rangle.$$

The integral is then

$$\int_C \nabla f \cdot d\mathbf{r} = f(2, 1, -1) - f(1, 1/2, 2)$$

$$= \cos(2\pi) + \sin(\pi) - 2(1)(-1) - \Big(\cos(\pi) + \sin(\pi/2) - 1(1/2)(2) \Big) = 4.$$

0.5.1 Independence of Path

Definition 3. *If* $\mathbf{F}$ *is a continuous vector field with domain* D, *we say that the line integral* $\int_C \mathbf{F} \cdot d\mathbf{r}$ *is* **independent of path** *if* $\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{C_2} \mathbf{F} \cdot d\mathbf{r}$ *for any two paths* C_1 *and* C_2 *in* D *that have the same initial and terminal points.*

With this terminology, we can say that line integrals of conservative vector fields are independent of path.

Theorem 3. $\int_C \mathbf{F} \cdot d\mathbf{r}$ *is independent of path in* D *if and only if* $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ *for every closed path* C *in* D.

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0.5.2 Some Useful Definitions

Suppose that **F** is a continuous vector field in some domain D.

Definition 4. **F** *is a* **conservative** *vector field if there is a function* f *such that* $\mathbf{F} = \nabla f$. *The function* f *is called a* **potential function** *for the vector field.*
Definition 5. *A path is called* **closed** *if its terminal point coincides with its initial point. For example, a circle is a closed path.*
Definition 6. *A path* C *is* **simple** *if it doesn't cross itself.*
Definition 7. *A region* D *is* **open** *if it doesn't contain any of its boundary points.*
Definition 8. *A region* D *is* **connected** *if we can connect any two points in the region with a path that lies completely in* D.
Definition 9. *A region* D *is* **simply-connected** *if it is connected and it contains no holes.*

0.5.3 Some Useful Theorems

Theorem 4. *Suppose* **F** *is a vector field that is continuous on an open connected region* D. *If* $\int_C \mathbf{F} \cdot d\mathbf{r}$ *is independent of path in* D, *then* **F** *is a conservative vector field on* D; *that is, there exists a function* f *such that* $\nabla f = \mathbf{F}$.

Theorem 5. *If* $\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$ *is a conservative vector field, where* P *and* Q *have continuous first-order partial derivatives on a domain* D, *then throughout* D *we have*

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

Theorem 6. *Let* $\mathbf{F}(x, y) = P\mathbf{i} + Q\mathbf{j}$ *be a vector field on an open simply-connected region* D. *Suppose that* P *and* Q *have continuous first-order partial derivatives*

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

throughout D. *Then* **F** *is conservative.*
Example 22. *Determine whether or not the following vector fields*

$$(a) \quad \mathbf{F}(x, y) = (3 + 2xy)\mathbf{i} + (x^2 - 3y^2)\mathbf{j}$$

$$(b) \quad \mathbf{F}(x, y) = (x^2 - xy)\mathbf{i} + (y^2 - xy)\mathbf{j}$$

are conservative.

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Solution:
(a) Let $P(x,y) = 3 + 2xy$ and $Q(x,y) = x^2 - 3y^2$. Then

$$\frac{\partial P}{\partial y} = 2x = \frac{\partial Q}{\partial x}.$$

The two partial derivatives are equal and so this is a conservative vector field.
(b) Let $P(x,y) = x^2 - xy$ and $Q(x,y) = y^2 - xy$. Then

$$\frac{\partial P}{\partial y} = -x \quad \textit{and} \quad \frac{\partial Q}{\partial x} = -y.$$

So, since the two partial derivatives are not the same this vector field is NOT conservative.

Suppose it is known that $\mathbf{F}(x,y) = P\mathbf{i} + Q\mathbf{j}$ is conservative, where P and Q have continuous first-order partial derivatives. Then there is a function f such that $\mathbf{F} = \nabla f$, that is,

$$P = \frac{\partial f}{\partial x} \quad \textit{and} \quad Q = \frac{\partial f}{\partial y}.$$

Example 23. *Determine if the following vector field is conservative and find a potential function for the vector field*

$$\mathbf{F}(x,y) = (2x^3y^4 + x)\mathbf{i} + (2x^4y^3 + y)\mathbf{j}$$

Solution:

$$P = 2x^3y^4 + x, \quad \frac{\partial P}{\partial y} = 8x^3y^3$$

and

$$Q = 2x^4y^3 + y, \quad \frac{\partial Q}{\partial x} = 8x^3y^3.$$

So, the vector field is conservative. Now to find the potential function, we know that

$$\frac{\partial f}{\partial x} = 2x^3y^4 + x, \quad \frac{\partial f}{\partial y} = 2x^4y^3 + y$$

From these we have

$$f(x,y) = \int \big(2x^3y^4 + x\big)dx \quad \textit{or} \quad f(x,y) = \int \big(2x^4y^3 + y\big)dy$$

Here is the first integral.

$$f(x,y) = \int \big(2x^3y^4 + x\big)dx$$

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$$= \frac{1}{2}x^4y^4 + \frac{1}{2}x^2 + h(y),$$

where $h(y)$ is the "constant of integration".
Differentiating f (including the $(h(y))$ with respect to y and set it equal to Q since that is what the derivative is supposed to be. Thus

$$\frac{\partial f}{\partial y} = 2x^4y^3 + h'(y) = 2x^4y^3 + y = Q$$

from which $h'(y) = y$. Thus

$$h(y) = \int h'(y)dy = \int ydy = \frac{1}{2}y^2 + c$$

So, putting this all together we can see that a potential function for the vector field is

$$f(x,y) = \frac{1}{2}x^4y^4 + \frac{1}{2}x^2 + \frac{1}{2}y^2 + c.$$

Example 24. *Evaluate* $\int_C \mathbf{F}\cdot d\mathbf{r}$, *where* $\mathbf{F} = (2x^3y^4 + x)\mathbf{i} + (2x^4y^3 + y)\mathbf{j}$ *and* C *is given by* $\mathbf{r(t)} = (\mathbf{t}\cos(\pi\mathbf{t}) - \mathbf{1})\mathbf{i} + \sin\left(\frac{\pi\mathbf{t}}{\mathbf{2}}\right)\mathbf{j}$, $0 \le t \le 1$.

Solution: From Example 23 above, this vector field is conservative and that a potential function for the vector field is

$$f(x,y) = \frac{1}{2}x^4y^4 + \frac{1}{2}x^2 + \frac{1}{2}y^2 + c.$$

Using this we know that integral must be independent of path and so all we need to do is use the **Fundamental Theorem for Line Integrals** to do the evaluation.
Thus

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int_C \nabla f\cdot d\mathbf{r} = f(\mathbf{r}(1)) - f(\mathbf{r}(0)),$$

where $\mathbf{r}(1) = \langle -2, 1\rangle$, $\mathbf{r}(0) = \langle -1, 0\rangle$.
So, the integral is

$$\int_C \mathbf{F}\cdot d\mathbf{r} = f(-2,1) - f(-1,0)$$

$$= \left(\frac{21}{2} + c\right) - \left(\frac{1}{2} + c\right) = 10.$$

Example 25. *If* $\mathbf{F}(x,y,z) = y^2\mathbf{i} + (2xy + e^{3z})\mathbf{j} + 3ye^{3z}\mathbf{k}$, *find a function* f *such that* $\nabla f = \mathbf{F}$.

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Solution: If there is such a function f, then

$$f_x(x, y, z) = y^2, \tag{12}$$

$$f_y(x, y, z) = 2xy + e^{3z}, \tag{13}$$

$$f_z(x, y, z) = 3ye^{3z}. \tag{14}$$

Integrating (12) with respect to x, we get

$$f(x, y, z) = xy^2 + g(y, z), \tag{15}$$

where $g(y, z)$ is a constant with respect to x. Then differentiating (15) with respect to y, we have

$$f_y(x, y, z) = 2xy + g_y(y, z) \tag{16}$$

and comparison with (13) gives

$$g_y(y, z) = e^{3z}.$$

Thus

$$g(y, z) = ye^{3z} + h(z)$$

and we rewrite (15) as

$$f(x, y, z) = xy^2 + ye^{3z} + h(z).$$

Finally, differentiating with respect to z and comparing with (14), we obtain $h'(z) = 0$ and therefore $h(z) = K$, a constant. The desired function is

$$f(x, y, z) = xy^2 + ye^{3z} + K.$$

EXERCISES

(a) Determine whether or not $\mathbf{F}$ s a conservative vector field. If it is, find a function f such that $\mathbf{F} = \nabla f$.

(i) $\mathbf{F}(x, y) = (ye^x + \sin y)\mathbf{i} + (e^x + x\cos y)\mathbf{j}$

(ii) $\mathbf{F}(x, y) = (2x - 3y)\mathbf{i} + (-3x + 4y - 8)\mathbf{j}$

(iii) $\mathbf{F}(x, y) = (2xy + y^{-2})\mathbf{i} + (x^2 - 2xy^{-3})\mathbf{j}, \quad y > 0.$

(b) Find a function f such that $\mathbf{F} = \nabla f$ and use the result to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the curve C.

(i) $\mathbf{F}(x, y) = x^2\mathbf{i} + y^2\mathbf{j}$, C is the arc of the parabola $y = 2x^2$ from $(-1, 2)$ to $(2, 8)$.

(ii) $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + (xy + 2z)\mathbf{k}$, C is the line segment from $(1, 0, -2)$ to $(4, 6, 3)$

(iii) $\mathbf{F}(x, y, z) = yze^{xz}\mathbf{i} + e^{xz}\mathbf{j} + xye^{xz}\mathbf{k}$, $C : \mathbf{r}(t) = (t^2 + 1)\mathbf{i} + (t^2 - 1)\mathbf{j} + (t^2 - 2t)\mathbf{k}$, $0 \le t \le 2$.

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(iv) $\mathbf{F}(x,y,z) = \sin y\mathbf{i} + (x\cos y + \cos z)\mathbf{j} - y\sin z\mathbf{k}$, $C : \mathbf{r}(t) = \sin t\mathbf{i} + t\mathbf{j} + 2t\mathbf{k}$, $0 \le t \le \frac{\pi}{2}$.

(c) Let

$$\mathbf{F}(x,y) = \frac{-y\mathbf{i} + x\mathbf{j}}{x^2 + y^2}$$

(i) Show that

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}.$$

(ii) Show that $\int_C \mathbf{F} \cdot d\mathbf{r}$ is not independent of path. [Hint: Compute $\int_{C_1} \mathbf{F} \cdot d\mathbf{r}$ and $\int_{C_2} \mathbf{F} \cdot d\mathbf{r}$, where C_1 and C_2 are the upper and lower halves of the circle $x^2 + y^2 = 1$ from $(1,0)$ to $(-1,0)$.]

(d) Let $\mathbf{F} = (ax^2y + y^3 + 1)\mathbf{i} + (2x^3 + bxy^2 + 2)\mathbf{j}$ be a vector field, where a and b are constants.

(i) Find the values of a and b for which $\mathbf{F}$ is conservative.

(ii) For these values of a and b, find $f(x,y)$ such that $\mathbf{F} = \nabla f$.

(iii) Still using the values of a and b from part (i), compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the curve C such that $x = e^t \cos t$, $y = e^t \sin t$, $0 \le t \le \pi$.

0.6 Green's Theorem

Theorem 7. *Let C be a positively oriented, piecewise-smooth, simple closed curve in the plane and let D be the region bounded by C. If P and Q have continuous partial derivatives on an open region that contains D, then*

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA.$$

Another notation for the positively oriented boundary curve of D is ∂D, so the equation in Green's Theorem can be written as

$$\int\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \int_{\partial D} P\,dx + Q\,dy. \tag{17}$$

Green's Theorem should be regarded as the counterpart of the Fundamental Theorem of Calculus for double integrals.

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Proof. Green's Theorem will be proved if we can show that

$$\int_C P dx = -\int\int_D \frac{\partial P}{\partial y} dA \tag{18}$$

$$\int_C Q dy = \int\int_D \frac{\partial Q}{\partial x} dA. \tag{19}$$

We prove Equation (18) by expressing D as a type 1 region:

$$D = \Big\{(x,y)|\ a \le x \le b,\ \ g_1(x) \le y \le g_2(x)\Big\},$$

where g_1 and g_2 are continuous functions. Thus

$$\int\int_D \frac{\partial P}{\partial y}\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y}(x,y)\, dy\, dx = \int_a^b \big[P(x, g_2(x)) - P(x, g_1(x))\big]\, dx \tag{20}$$

Now we compute the left side of Equation (18) by breaking up C as the union of the four curves $C_1,\ C_2,\ C_3$ *and* C_4 shown in Figure 8.

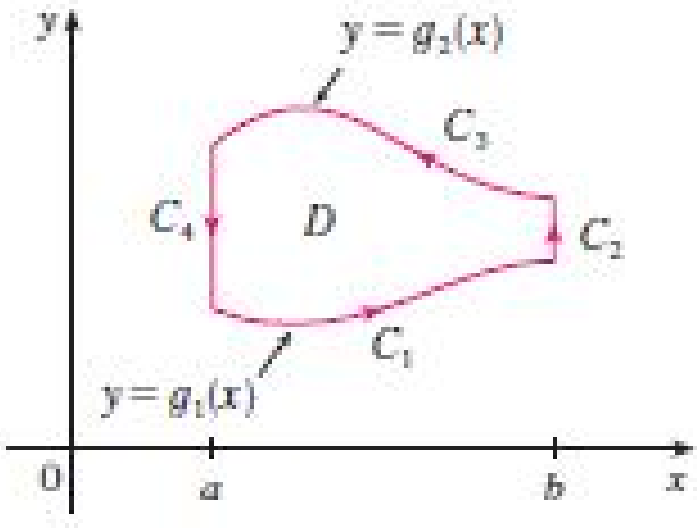


Figure 26:

On C_1, we take x as the parameter and write the parametric equations as $x = x,\ y = g_1(x),\ a \le x \le b$. Thus

$$\int_{C_1} P(x,y) dx = \int_a^b P(x, g_1(x))$$

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Since C_3 goes from right to left but $-C_3$ goes from left to right, so we can write the parametric equation $-C_3$ as $x = x,\ y = g_2(x),\ a \leq x \leq b$. Therefore

$$\int_{C_3} P(x,y)dx = -\int_{C_3} P(x,y)dx = -\int_a^b P(x, g_2(x)).$$

On C_2 or C_4, x is constant, so $dx = 0$ and

$$\int_{C_2} P(x,y)dx = 0 = \int_{C_4} P(x,y)dx.$$

Hence

$$\int_C P(x,y)dx = \int_{C_1} P(x,y)dx + \int_{C_2} P(x,y)dx + \int_{C_3} P(x,y)dx + \int_{C_4} P(x,y)dx.$$

$$= \int_a^b P(x, g_2(x))dx - \int_a^b P(x, g_1(x))dx.$$

Comparing this expression with the one in Equation (20), we see that

$$\int_C Pdx = -\int\int_D \frac{\partial P}{\partial y} dA.$$

□

Example 26. *Evaluate* $\int_C x^4 dx + xydy$, *where* C *is the triangular curve consisting of the line segments from* $(0,0)$ *to* $(1,0)$, *from* $(1,0)$ *to* $(0,1)$, *and from* $(0,1)$ *to* $(0,0)$.

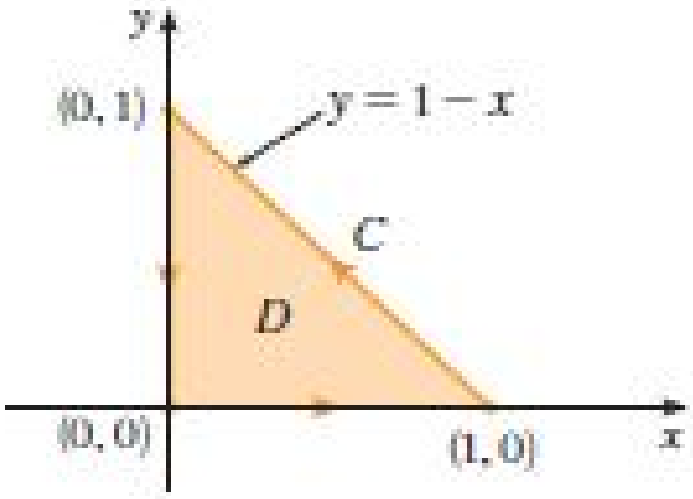


Figure 27:

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Solution: The region D enclosed by C is simple and C has a positive orientation (see Figure 27).
If we let $P(x,y) = x^4$ and $Q(x,y) = xy$, then we have

$$\int_C x^4 dx + xy dy = \int\int_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA = \int_0^1 \int_0^{1-x} (y-0)\, dy\, dx$$

$$= \int_0^1 \left[\frac{1}{2}y^2\right]_{y=0}^{y=1-x} dx = \frac{1}{2}\int_0^1 (1-x)^2 dx$$

$$= -\frac{1}{6}(1-x)^3\Big|_0^1 = \frac{1}{6}.$$

Example 27. *Evaluate* $\oint_C (3y - e^{\sin x})dx + \left(7x + \sqrt{y^4+1}\right)dy$, *where* C *is the circle* $x^2+y^2 = 9$.

Solution: The region D bounded by C is the disk $x^2 + y^2 \leq 9$. So, using polar coordinates after applying Greens Theorem, we have

$$\oint_C (3y - e^{\sin x})\, dx + \left(7x + \sqrt{y^4+1}\right) dy$$

$$= \int\int_D \left[\frac{\partial}{\partial x}\left(7x + \sqrt{y^4+1}\right) - \frac{\partial}{\partial y}\left(3y - e^{\sin x}\right)\right] dA$$

$$= \int_0^{2\pi}\int_0^3 (7-3) r\, dr\, d\theta = 4\int_0^{2\pi} d\theta \int_0^3 r\, dr = 36\pi.$$

The Green's Theorem gives the following formulas for the area of D

$$A = \oint_C x\, dy = -\oint_C y\, dx = \frac{1}{2}\oint_C x\, dy - y\, dx. \tag{21}$$

Example 28. *Find the area enclosed by the ellipse* $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Solution: The ellipse has parametric equations $x = a\cos t$ and $y = b\sin t$, where $0 \leq t \leq 2\pi$.
Using the third formula in Equation (21), we have

$$A = \frac{1}{2}\oint_C x\, dy - y\, dx$$

$$= \frac{1}{2}\int_0^{2\pi} (a\cos t)(b\cos t)\, dt - (b\sin t)(-a\sin t)\, dt$$

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$$= \frac{ab}{2} \int_0^{2\pi} dt = \pi ab.$$

EXERCISES

(1) Evaluate the line integral by two methods: (a) directly and (b) using Green's Theorem.

(i) $\oint_C (x - y)\, dx + (x + y)\, dy$; C is the circle with center the origin and radius 2.

(ii) $\oint_C xy\, dx + x^2y^3\, dy$; C is the triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$.

(iii) $\oint_C x^2y^2\, dx + xy\, dy$; C consists of the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$ and the line segments from $(1, 1)$ to $(0, 1)$ and $(0, 1)$ to $(0, 0)$.

(2) Use Green's Theorem to evaluate the line integral along the given positively oriented curve.

(i) $\int_C xy^2\, dx + 2x^2y\, dy$; C is the triangle with vertices $(0, 0)$, $(2, 2)$ and $(2, 4)$.

(ii) $\int_C (y + e^{\sqrt{x}})\, dx + (2x + \cos y^2)\, dy$; C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$.

(iii) $\int_C (1 - y^3)\, dx + (x^3 + e^{y^2})\, dy$; C is the boundary of the region between the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$.

(3) Use Green's Theorem to find the work done by the force $\mathbf{F}(x, y) = x(x + y)\mathbf{i} + xy^2\mathbf{j}$ in moving a particle from the origin along the x-axis to $(1, 0)$, then along the line segment to $(0, 1)$, and then back to the origin along the y-axis.

(4) A particle starts at the point $(-2, 0)$, moves along the x-axis to $(2, 0)$, and then along the semicircle $y = \sqrt{4 - x^2}$ to the starting point. Use Green's Theorem to find the work done on this particle by the force field $\mathbf{F}(x, y) = \langle x, x^3 + 3xy^2 \rangle$.

0.7 Stokes' Theorem

Stokes' Theorem: Let S be an oriented piecewise-smooth surface that is bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation. Let $\mathbf{F}$ be a vector field whose components have continuous partial derivatives on an open region in $\mathbb{R}^3$ that contains S. Then

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int \int_S curl\, \mathbf{F} \cdot\, d\mathbf{S}.$$

Note:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T}\, ds \text{ and } \int \int_S curl\, \mathbf{F} \cdot\, d\mathbf{S} = \int \int_S curl\, \mathbf{F} \cdot \mathbf{n}\, dS.$$

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Stokes Theorem says that the line integral around the boundary curve of S of the tangential component of $\mathbf{F}$ is equal to the surface integral over S of the normal component of the curl of $\mathbf{F}$.

The positively oriented boundary curve of the oriented surface S is often written as ∂S, so Stokes' Theorem can be expressed as

$$\int\int_S curl\,\mathbf{F}\cdot\,d\mathbf{S} = \int_{\partial S}\mathbf{F}\cdot d\mathbf{r} \tag{22}$$

In the special case where the surface S is flat and lies in the xy-plane with upward orientation, the unit normal is $\mathbf{k}$, the surface integral becomes a double integral, and Stokes' Theorem becomes

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int\int_S curl\,\mathbf{F}\cdot\,d\mathbf{S} = \int\int_S (curl\,\mathbf{F})\cdot\mathbf{k}\,dA.$$

This is precisely the vector form of Green's Theorem.

Example 29. *Evaluate* $\int_C \mathbf{F}\cdot d\mathbf{r}$, *where* $\mathbf{F}(x,y,z) = -y^2\mathbf{i} + x\mathbf{j} + z^2\mathbf{k}$ *and* C *is the curve of intersection of the plane* $y+z=2$ *and the cylinder* $x^2+y^2=1$. *(Orient* C *to be counterclockwise) when viewed from above.*

Solution: The curve C (an ellipse) is shown in Figure 28.

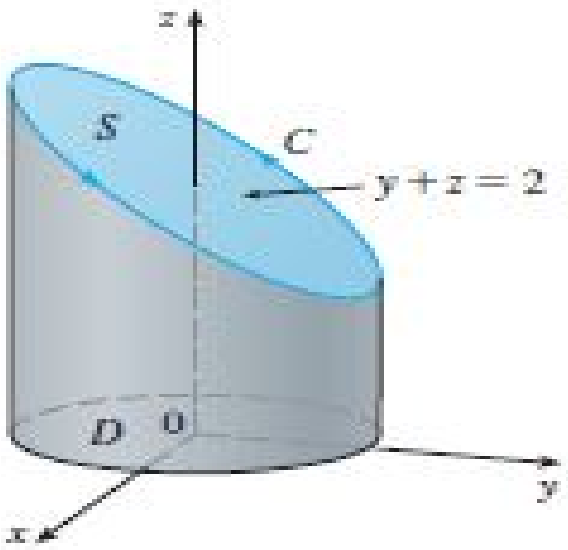


Figure 28:

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Using **Stokes' Theorem,** we have

$$curl\,\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y^2 & x & z^2 \end{vmatrix} = (1+2y)\underline{\mathrm{k}}.$$

Although there are many surfaces with boundary C, the most convenient choice is the elliptical region S in the plane $y+z=2$ that is bounded by C. If we orient S upward, then C has the induced positive orientation. The projection D of S onto the xy-plane is the disk $x^2+y^2 \le 1$.
Thus we have

$$\int_C \mathbf{F}\cdot d\mathbf{r} = \int\int_S curl\,\mathbf{F}\cdot\, d\mathbf{S} = \int\int_D (1+2y)\,dA.$$

$$= \int_0^{2\pi}\int_0^1 (1+2r\sin\theta) r\,dr\,d\theta = \int_0^{2\pi}\left[\frac{r^2}{2}+2\frac{r^3}{3}\sin\theta\right]_0^1 d\theta$$

$$= \int_0^{2\pi}\left(\frac{1}{2}+\frac{2}{3}\sin\theta\right)d\theta = \frac{1}{2}(2\pi)+0=\pi.$$

Example 30. *Use Stokes' Theorem to compute the integral* $\int\int_S curl\,\mathbf{F}\cdot d\mathbf{S}$, *where* $\mathbf{F}(x,y,z) = xz\mathbf{i}+yz\mathbf{j}+xy\mathbf{k}$ *and* S *is the part of the sphere* $x^2+y^2+z^2=4$ *that lies inside the cylinder* $x^2+y^2=1$ *and above the* xy*-plane. (See Figure* 29*).*

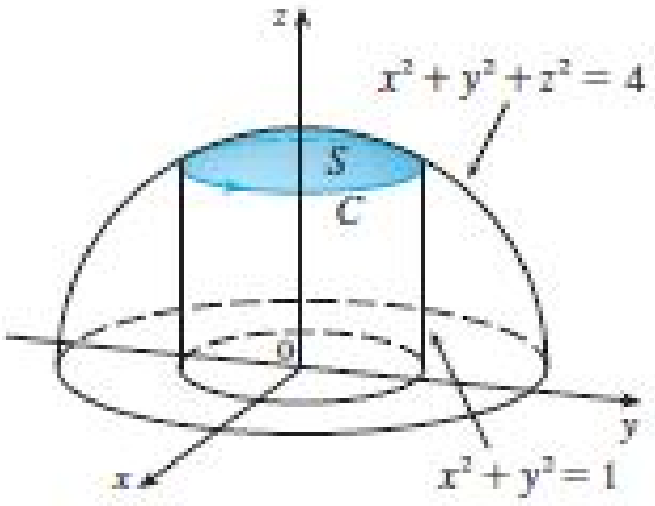


Figure 29:

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Solution: To find the boundary curve C, we solve the equations $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 = 1$. Subtracting, we get $z^2 = 3$ and so $z = \sqrt{3}$ (*since* $z > 0$). Thus C is the circle given by the equation $x^2 + y^2 = 1$, $z = \sqrt{3}$.
A vector equation of C is therefore,

$$\mathbf{r}(t) = \cos\, t\mathbf{i} + \sin\, t\mathbf{j} + \sqrt{3}\mathbf{k}, \quad 0 \le t \le 2\pi,$$

so

$$\mathbf{r}(t) = -\sin\, t\mathbf{i} + \cos\, t\mathbf{j}.$$

Also, we have

$$\mathbf{F}(\mathbf{r}(t)) = \sqrt{3}\cos\, t\,\mathbf{i} + \sqrt{3}\sin\, t\,\mathbf{j} + \cos\, t \sin\, t\,\mathbf{k}$$

Therefore, by Stokes' Theorem,

$$\int\int_S curl\,\mathbf{F} \cdot\, d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$$

$$= \int_0^{2\pi} \Big(-\sqrt{3}\cos\, t\, \sin\, t + \sqrt{3}\sin\, t\, \cos\, t \Big)\, dt$$

$$= \sqrt{3}\int_0^{2\pi} 0\, dt = 0.$$

In this example, we computed a surface integral simply by knowing the values of $\mathbf{F}$ on the boundary curve C. This means that if we have another oriented surface with the same boundary curve C, then we get exactly the same value for the surface integral!

In general, if S_1 and S_2 are oriented surfaces with the same oriented boundary curve C and both satisfy the hypotheses of Stokes' Theorem, then

$$\int\int_{S_1} curl\,\mathbf{F} \cdot\, d\mathbf{S} = \int_C \mathbf{F} \cdot d\mathbf{r} = \int\int_{S_2} curl\,\mathbf{F} \cdot\, d\mathbf{S} \tag{23}$$

This fact is useful when it is difficult to integrate over one surface but easy to integrate over the other.

EXERCISES
(a) Use Stokes Theorem to evaluate $\int\int_S curl\,\mathbf{F} \cdot\, d\mathbf{S}$

(i) $\mathbf{F}(x, y, z) = 2y\cos\, z\mathbf{i} + e^x \sin\, z\mathbf{j} + xe^y\mathbf{k}$, S is the hemisphere $x^2 + y^2 + z^2 = 9$, $z \ge 0$, oriented upward.

(ii) $\mathbf{F}(x, y, z) = x^2z^2\,\mathbf{i} + y^2z^2\,\mathbf{j} + xyz\,\mathbf{k}$, S is the part of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 4$, oriented upward.

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(iii) $\mathbf{F}(x,y,z) = tan^{-1}(x^2yz^2)\,\mathbf{i} + x^2y\,\mathbf{j} + x^2z^2\,\mathbf{k}$, S is the cone $x = \sqrt{y^2+z^2}$, $0 \le x \le 2$, oriented in the direction of the positive x-axis.

(b) Use Stokes' Theorem to evaluate $\int_C \mathbf{F}\cdot d\mathbf{r}$. In each case C is oriented counterclockwise as viewed from above

(i) $\mathbf{F}(x,y,z) = (x+y^2)\,\mathbf{i} + (y+z^2)\,\mathbf{j} + (z+x^2)\mathbf{k}$, C is the triangle with vertices $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$.

(ii) $\mathbf{F}(x,y,z) = yz\,\mathbf{i} + 2xz\,\mathbf{j} + e^{xy}\,\mathbf{k}$, C is the circle $x^2+y^2 = 16$, $z = 5$..

(iii) $\mathbf{F}(x,y,z) = xy\,\mathbf{i} + 2z\,\mathbf{j} + 3y\,\mathbf{k}$, C is the curve of intersection of the plane $x+z = 5$ and the cylinder $x^2+y^2 = 9$.

0.8 The Divergence Theorem

The Divergence Theorem: Let E be a simple solid region and let S be the boundary surface of E, given with positive (outward) orientation. Let $\mathbf{F}$ be a vector field whose component functions have continuous partial derivatives on an open region that contains E. Then

$$\int\int_S \mathbf{F}\cdot\, d\mathbf{S} = \int\int\int_E div\,\mathbf{F}\, dV.$$

Thus the Divergence Theorem states that, under the given conditions, the flux of $\mathbf{F}$ across the boundary surface of E is equal to the triple integral of the divergence of $\mathbf{F}$ over E.

Example 31. *Use the divergence theorem to evaluate* $\int\int_S \mathbf{F}\cdot d\mathbf{S}$, *where* $\mathbf{F}(x,y,z) = xy\,\mathbf{i} - \frac{1}{2}y^2\,\mathbf{j} + z\mathbf{k}$ *and the surface consists of the three surfaces,* $z = 4-3x^2-3y^2$, $1 \le z \le 4$ *on the top* $x^2+y^2 = 1$, $0 \le z \le 1$ *on the sides and* $z = 0$ *on the bottom.*

Solution: The sketch of the surface is shown in Figure 30.

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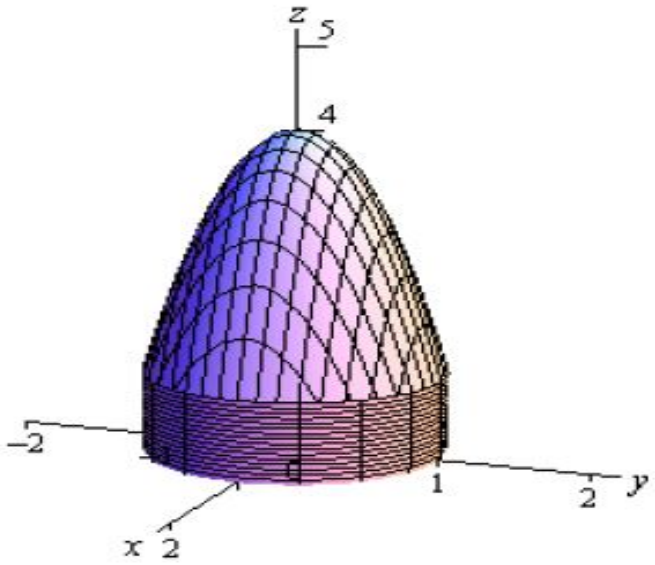


Figure 30:

The region E for the triple integral is then the region enclosed by these surfaces. Note that cylindrical coordinates would be a perfect coordinate system for this region. If we do that here are the limits for the ranges.

$$0 \le z \le 4 - 3r^2, \quad 0 \le r \le 1, \quad 0 \le \theta \le 2\pi.$$

We'll also need the divergence of the vector field, so we have

$$div\,\mathbf{F} = y - y + 1 = 1.$$

The integral is then,

$$\begin{aligned}
\int\int_S \mathbf{F} \cdot d\mathbf{S} &= \int\int\int_E div\,\mathbf{F}\, dV \\
&= \int_0^{2\pi}\int_0^1\int_0^{4-3r^2} r\, dz\, dr\, d\theta \\
&= \int_0^{2\pi}\int_0^1 (4r - 3r^3)\, dr\, d\theta \\
&= \int_0^{2\pi} \left(2r^2 - \frac{3}{4}r^4\right)\Big|_0^1 d\theta \\
&= \int_0^{2\pi} \frac{5}{4}\, d\theta = \frac{5}{2}\pi.
\end{aligned}$$

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Example 32. *Evaluate* $\int\int_S \mathbf{F} \cdot d\mathbf{S}$, *where*

$$\mathbf{F}(x, y, z) = xy\,\mathbf{i} + (y^2 + e^{xz^2})\,\mathbf{j} + \sin(xy)\,\mathbf{k}$$

and S *is the surface of the region* E *bounded by the parabolic cylinder* $z = 1 - x^2$ *and the planes* $z = 0$, $y = 0$, *and* $y + z = 2$. *(See Figure* 31*).*

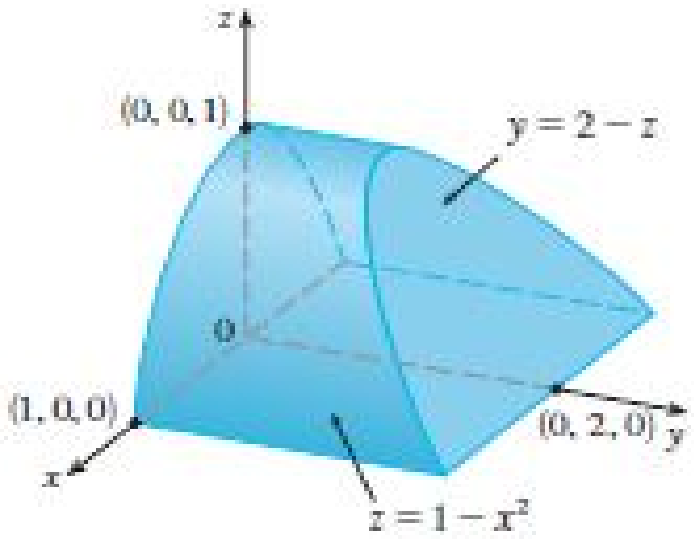


Figure 31:

Solution: The divergence of $\mathbf{F}$ is

$$div\,\mathbf{F} = \frac{\partial}{\partial x}\Big(xy\Big) + \frac{\partial}{\partial y}\Big(y^2 + e^{xz^2}\Big) + \frac{\partial}{\partial z}\Big(\sin(xy)\Big) = y + 2y = 3y$$

Therefore we use the Divergence Theorem to transform the given surface integral into a triple integral. The easiest way to evaluate the triple integral is to express E as a type 3 region:

$$E = \Big\{(x, y, z) |\ -1 \le x \le 1,\ 0 \le z \le 1 - x^2,\ 0 \le y \le 2 - z\Big\}$$

Then we have

$$\int\int_S \mathbf{F} \cdot d\mathbf{S} = \int\int\int_E div\,\mathbf{F}\,dV = \int\int\int_E 3y\,dV$$

$$= 3\int_{-1}^{1}\int_{0}^{1-x^2}\int_{0}^{2-z} y\,dy\,dz\,dx = 3\int_{-1}^{1}\int_{0}^{1-x^2}\frac{(2-z)^2}{2}\,dz\,dx$$

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$$= \frac{3}{2}\int_{-1}^{1}\left[-\frac{(2-z)^3}{3}\right]_0^{1-x^2} dx = -\frac{1}{2}\int_{-1}^{1}[(x^3+1)^3-8]\,dx$$

$$= -\int_0^1 \left(x^6+3x^4+3x^2-7\right) dx = \frac{184}{35}.$$

EXERCISES

(a) Use the Divergence Theorem to calculate the surface integral $\int\int_S \mathbf{F}\cdot d\mathbf{S}$; that is, calculate the flux of $\mathbf{F}$ over S.

(i) $\mathbf{F}(x,y,z) = xye^z\,\mathbf{i} + xy^2z^3\,\mathbf{j} - ye^z\,\mathbf{k}$, S is the surface of the box bounded by the coordinate planes and the planes $x=3$, $y=2$, and $z=1$.

(ii) $\mathbf{F}(x,y,z) = 3xy^2\,\mathbf{i} + xe^z\,\mathbf{j} + z^3\,\mathbf{k}$, S is the surface of the solid bounded by the cylinder $y^2+z^2=1$ and the planes $x=-1$ and $x=2$.

(iii) $\mathbf{F}(x,y,z) = (\cos z + xy^2)\,\mathbf{i} + xe^{-z}\,\mathbf{j} + (\sin y + x^2z)\,\mathbf{k}$, S is the surface of the solid bounded by the paraboloid $z=x^2+y^2$ and the plane $x=4$.

(iv) $\mathbf{F}(x,y,z) = (x^3+y^3)\,\mathbf{i} + (y^3+z^3)\,\mathbf{j} + (z^3+x^3)\,\mathbf{k}$, S is the sphere with center the origin and radius 2.

0.9 Improper Integral

The concept of a definite integral can be extended to the case where the interval is infinite and also to the case where f has an infinite discontinuity in $[a,b]$. In either case the integral is called an **improper integral**.

0.9.1 Type 1: Infinite Intervals

In this kind of integral one or both of the limits of integration are infinity. In these cases the interval of integration is said to be over an infinite interval.
Example 33. *Evaluate the following integral*

$$\int_1^\infty \frac{1}{x^2}\,dx.$$

Solution: Because infinity is not a real number, we can't just integrate the integral as normal and then "plug in" the infinity to get an answer.

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Consider when the area under $f(x)$ is on the interval $[1, t]$, where $t > 1$ and t is finite. Thus we have

$$A_t = \int_1^t \frac{1}{x^2}\,dx = -\frac{1}{x}\bigg|_1^t = 1 - \frac{1}{t}.$$

Now, we can get the area under $f(x)$ on $[1, \infty)$ simply by taking the limit of A_t as t goes to infinity.

$$A = \lim_{t\to\infty} A_t = \lim_{t\to\infty}\left(1 - \frac{1}{t}\right) = 1.$$

So we have

$$\begin{aligned}\int_1^\infty \frac{1}{x^2}\,dx &= \lim_{t\to\infty}\int_1^t \frac{1}{x^2}\,dx \\ &= \lim_{t\to\infty}\left(-\frac{1}{x}\right)\bigg|_1^t \\ &= \lim_{t\to\infty}\left(1 - \frac{1}{t}\right) = 1.\end{aligned}$$

Definition of an Improper Integral of Type 1

Definition 10. *(a) If $\int_a^t f(x)\,dx$ exists for every number $t \geq a$, then*

$$\int_a^\infty f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx$$

provided this limit exists (as a finite number).

(b) If $\int_t^b f(x)\,dx$ exists for every number $t \leq b$, then

$$\int_{-\infty}^b f(x)\,dx = \lim_{t\to-\infty}\int_t^b f(x)\,dx$$

provided this limit exists (as a finite number).
The improper integrals $\int_a^\infty f(x)\,dx$ and $\int_{-\infty}^b f(x)\,dx$ are called **convergent** *if the corresponding limit exists and* **divergent** *if the limit does not exist.*

(c) If both $\int_a^\infty f(x)\,dx$ and $\int_{-\infty}^a f(x)\,dx$ are convergent, then we define

$$\int_{-\infty}^\infty f(x)\,dx = \int_{-\infty}^a f(x)\,dx + \int_a^\infty f(x)\,dx$$

In part (c) any real number a can be used.

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Example 34. *Determine whether the integral* $\int_1^\infty (\frac{1}{x})\,dx$ *is convergent or divergent.*

Solution: According to part (a) of Definition 9, we have

$$\int_1^\infty \left(\frac{1}{x}\right) dx = \lim_{t\to\infty} \int_1^t \frac{1}{x}\,dx = \lim_{t\to\infty} \ln|x|\Big|_1^t$$

$$= \lim_{t\to\infty} (\ln t - \ln 1) = \lim_{t\to\infty} \ln t = \infty.$$

The limit does not exist as a finite number and so the improper integral $\int_1^\infty (\frac{1}{x})\,dx$ is divergent.

Example 35. *Evaluate*

$$\int_{-\infty}^0 xe^x\,dx$$

Solution: Using part (b) of Definition 9, we have

$$\int_{-\infty}^0 xe^x\,dx = \lim_{t\to-\infty} \int_t^0 xe^x\,dx.$$

We integrate by parts with $u = x$, $dv = e^x\,dx$ so that $du = dx$, $v = e^x$:

$$\int_t^0 xe^x\,dx = xe^x\Big|_t^0 - \int_t^0 e^x\,dx$$

$$= -te^t - 1 + e^t.$$

But $e^t \to 0$ as $t \to -\infty$, and by L'Hospital's Rule we have

$$\lim_{t\to-\infty} te^t = \lim_{t\to-\infty} \frac{t}{e^{-t}} = \lim_{t\to-\infty} \frac{1}{-e^{-t}}$$

$$= \lim_{t\to-\infty} (-e^{-t}) = 0.$$

Therefore

$$\int_{-\infty}^0 xe^x\,dx = \lim_{t\to-\infty} (-te^t - 1 + e^t)$$

$$= -0 - 1 + 0 = -1.$$

Example 36. *Evaluate*

$$\int_{-\infty}^\infty \frac{1}{1+x^2}\,dx$$

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Solution: Choosing $a = 0$ in Definition 9(*c*) :

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \int_{-\infty}^{0} \frac{1}{1+x^2}\,dx + \int_{0}^{\infty} \frac{1}{1+x^2}\,dx$$

We must now evaluate the integrals on the right side separately:

$$\int_{0}^{\infty} \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \int_{0}^{t} \frac{1}{1+x^2}\,dx = \lim_{t\to\infty} \tan^{-1} x \Big|_{0}^{t}$$

$$= \lim_{t\to\infty} (\tan^{-1} t - \tan^{-1} 0) = \lim_{t\to\infty} \tan^{-1} t = \frac{\pi}{2}.$$

$$\int_{-\infty}^{0} \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \int_{t}^{0} \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty} \tan^{-1} x \Big|_{t}^{0}$$

$$= \lim_{t\to-\infty} (\tan^{-1} 0 - \tan^{-1} t)$$

$$= 0 - \Big(-\frac{\pi}{2}\Big) = \frac{\pi}{2}$$

Since both of these integrals are convergent, the given integral is convergent and

$$\int_{-\infty}^{\infty} \frac{1}{1+x^2}\,dx = \frac{\pi}{2} + \frac{\pi}{2} = \pi.$$

0.9.2 Discontinuous Integrand

We now look at the second type of improper integrals.

Definition of an Improper Integral of Type 2

Definition 11.

(a) If f is continuous on $[a, b)$ and is discontinuous at b, then

$$\int_{a}^{b} f(x)\,dx = \lim_{t\to b^-} \int_{a}^{t} f(x)\,dx$$

if this limit exists (as a finite number).

(b) If f is continuous on $(a, b]$ and is discontinuous at a, then

$$\int_{a}^{b} f(x)\,dx = \lim_{t\to a^+} \int_{t}^{b} f(x)\,dx$$

if this limit exists (as a finite number).
The improper integrals $\int_a^b f(x)\,dx$ is called **convergent** *if the corresponding limit exists and* **divergent** *if the limit does not exist.*

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(c) If f has a discontinuity at c, where $a < c < b$, and both $\int_a^c f(x)\,dx$ and $\int_c^b f(x)\,dx$ are convergent, then we define

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

Example 37. *Find*

$$\int_2^5 \frac{1}{\sqrt{x-2}}\,dx$$

Solution: We note first that the given integral is improper because $f(x) = \frac{1}{\sqrt{x-2}}$ has the vertical asymptote $x = 2$. Since the infinite discontinuity occurs at the left endpoint of $[2, 5]$, we use we use part (b) of Definition 10 :

$$\begin{aligned}\int_2^5 \frac{1}{\sqrt{x-2}}\,dx &= \lim_{t\to 2^+}\int_t^5 \frac{1}{\sqrt{x-2}}\,dx \\ &= \lim_{t\to 2^+} 2\sqrt{x-2}\Big|_t^5 \\ &= \lim_{t\to 2^+} 2\big(\sqrt{3}-\sqrt{t-2}\big) = 2\sqrt{3}.\end{aligned}$$

Example 38. *Evaluate*

$$\int_0^3 \frac{1}{x-1}\,dx$$

if possible

Solution: Observe that the line $x = 1$ is a vertical asymptote of the integrand. Since it occurs in the middle of the interval $[0, 3]$, we must use part (c) of Definition 10 with $c = 1$:

$$\int_0^3 \frac{1}{x-1}\,dx = \int_0^1 \frac{1}{x-1}\,dx + \int_1^3 \frac{1}{x-1}\,dx,$$

where

$$\begin{aligned}\int_0^1 \frac{1}{x-1}\,dx = \lim_{t\to 1^-}\int_0^t \frac{1}{x-1}\,dx &= \lim_{t\to 1^-} \ln|x-1|\Big|_0^t \\ &= \lim_{t\to 1^-}(\ln|t-1| - \ln|-1|) \\ &= \lim_{t\to 1^-} \ln(1-t) = -\infty\end{aligned}$$

because $1 - t \to 0^+$ as $t \to 1^-$. Thus $\int_0^1 \frac{1}{x-1}\,dx$ is divergent. This implies that $\int_0^3 \frac{1}{x-1}\,dx$ is divergent.

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EXERCISES

(a) Determine whether each integral is convergent or divergent. Evaluate those that are convergent.

$$\int_3^\infty \frac{1}{(x-2)^{3/2}}\,dx, \qquad \int_{-\infty}^0 \frac{1}{3-4x}\,dx, \qquad \int_{-\infty}^\infty xe^{-x^2}\,dx, \qquad \int_1^\infty \frac{\ln x}{x}\,dx.$$

(b) The integral

$$\int_0^\infty \frac{1}{\sqrt{x}(1+x)}\,dx$$

is improper for two reasons: The interval $[0, \infty)$ is infinite and the integrand has an infinite discontinuity at 0. Evaluate it by expressing it as a sum of improper integrals of Type 2 and Type 1 as follows:

$$\int_0^\infty \frac{1}{\sqrt{x}(1+x)}\,dx = \int_0^1 \frac{1}{\sqrt{x}(1+x)}\,dx + \int_1^\infty \frac{1}{\sqrt{x}(1+x)}\,dx$$

(c) Evaluate

$$\int_2^\infty \frac{1}{x\sqrt{x^2-4}}\,dx$$

by the same method as in Exercise *b*.

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